A Candidate Four-Derivative Causal-Set Operator

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The Einstein-Hilbert action is the leading term in the derivative expansion of gravitational dynamics. Its four-derivative completion is central both to effective field theory and to approaches to ultraviolet quantum gravity. We study a finite-layer causal-set ansatz intended to represent a fourth-order scalar operator and its associated geometric action contribution. In four spacetime dimensions, the layer polynomial is supported on six past causal layers, with coefficients $(1, -24, 343/3, -214, 176, -160/3)$. The Poisson-averaged continuum analysis leads to a finite set of local candidate terms, but the complete covariant reduction has not yet been performed. In the 30-term intermediate expansion, only the first nine structures are manifest spacetime scalars; the remaining terms involve components relative to a distinguished time direction and must be reduced after the full summation. The normalization coefficients α_d and β_d, together with the complete curvature dressing of the continuum operator, therefore remain to be determined. We present the finite-layer construction and the framework for its continuum analysis, and discuss its possible relation to curvature-squared effective gravity, locality, boundary terms, and a hierarchy of higher-derivative causal-set contributions.

Introduction. Gravity admits a systematic organization in derivatives of the metric. At the lowest order, the Einstein-Hilbert term governs the long-distance dynamics; at the next nontrivial order, the local action contains operators quadratic in curvature, schematically R^2, $R_{\mu\nu}R^{\mu\nu}$ and $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$, together with derivative terms and boundary contributions. These operators are unavoidable in the gravitational effective field theory and play an important role in ultraviolet descriptions of gravity [1–3]. In four dimensions they have the crucial engineering dimension of a Lagrangian density: $[R^2] = L^{-4}$, so that $d^4x\sqrt{-g}R^2$ is dimensionless in units with $\hbar = c = 1$. A microscopic theory of spacetime should therefore not merely reproduce curvature as an emergent observable, but should provide a natural microscopic representative of this next order in the gravitational action.

Causal set theory takes local finiteness and causal order as fundamental, with cardinality playing the role of spacetime volume [4, 5]. This structure already supports a remarkable action principle. Retarded finite-layer operators on a causal set reproduce the continuum scalar d'Alembertian in appropriate continuum limits, and the same construction yields the Ricci scalar and hence the Benincasa-Dowker gravitational action [6–8]. The resulting random discrete action has a direct continuum interpretation in terms of the Einstein-Hilbert action, including boundary effects that reflect the bilocal character of the microscopic construction [9, 10]. The natural question at the next derivative order is then not only whether one can identify additional curvature invariants, but whether the causal order itself can furnish their action density.

This question is also relevant beyond causal set theory. In continuum quantum gravity, a four-derivative truncation introduces new dimensionless couplings and is closely tied to questions of ultraviolet completion and renormalization-group flow [2, 3]. In causal set theory, higher-curvature operators have previously been obtained by iterating the discrete d'Alembertian; in particular, the continuum image of a B^2 construction contains a combination of R^2 and $\Box R$ [11]. Here we take a different route: the four-derivative term is itself represented by one finite causal kernel. This makes the fourth-order contribution an independent microscopic building block rather than a composite of lower-order operators.

A Four-Derivative Causal Density. Let $L_i(x)$ denote the ith past layer of a causal-set element x. We consider the retarded operator

$$\mathcal{B}_4^{(4)}\phi(x) = \frac{1}{\ell^4}\left[\alpha_4\phi(x) + \beta_4\sum_{i=1}^{6} C_i \sum_{y\in L_i(x)} \phi(y)\right], \quad (1)$$

with

$$(C_1, C_2, C_3, C_4, C_5, C_6) = \big(1, -24, \frac{343}{3}, -214, 176, -\frac{160}{3}\big). \quad (2)$$

The associated layer polynomial is

$$P_4(z) = 1 - 24z + \frac{343}{6}z^2 - \frac{107}{3}z^3 + \frac{22}{3}z^4 - \frac{4}{9}z^5. \quad (3)$$

The constants α_4 and β_4 are to be fixed by the continuum normalization conditions. The former cancels the leading density-dependent contribution proportional to $\phi(x)$, while the latter is fixed by requiring the coefficient of $\Box^2\phi$ in the finite local limit to be unity. Since the complete summation and covariant reduction in Appendix A remain to be completed, we do not quote their numerical values here. The layer polynomial itself is fixed, but the fully normalized continuum operator has not yet been established.

The six-layer structure is the four-dimensional member of a general-dimensional family. For integer spacetime dimension $d \geq 2$, the fourth-order construction requires

$$n_d = \left\lfloor \frac{d}{2} \right\rfloor + 4 \tag{4}$$

causal layers. Thus the microscopic range of the operator is finite at every fixed dimension and is determined by the derivative order rather than introduced as an independent nonlocality scale. The general construction and the dimension-dependent continuum analysis are given in Appendix A; the geometric input from the Alexandrov volume is derived in Appendix C.

Continuum Limit. For a Poisson sprinkling of density $\rho = \ell^{-4}$ into a smooth four-dimensional Lorentzian spacetime, the aim is to determine the local continuum limit of the expectation value of the finite-layer operator.

The expected covariant form of the limiting operator is

$$\mathcal{D}_4\phi = \sum_{A=1}^{9} \gamma_A \mathcal{K}_A, \qquad \gamma_2 = 1, \qquad \gamma_1 = 0, \tag{5}$$

where $\mathcal{K}_2 = \Box^2\phi$ and $\mathcal{K}_1, \ldots, \mathcal{K}_9$ are the scalar structures listed in Appendix A. The coefficient γ_2 fixes the normalization of the fourth-derivative term, while $\gamma_1 = 0$ represents cancellation of the standalone ϕ contribution. The remaining coefficients have not yet been evaluated.

The intermediate calculation contains 30 contractions, of which only the first nine are manifest spacetime scalars. The remaining contractions depend on the distinguished time direction of the adapted frame. Lorentz covariance requires the fully summed result to reduce to a scalar expression, but does not by itself establish that each frame-dependent contraction vanishes separately. The complete summation and tensor reduction are needed to determine which combinations survive and to fix their coefficients.

Accordingly, Eq. (5) specifies the target covariant structure rather than a completed evaluation of the continuum limit. The remaining calculation will determine the curvature and derivative corrections and establish the full local operator.

The intermediate expressions also retain the geometric information carried by the Alexandrov interval. Their decomposition relative to the timelike direction associated with the causal interval is useful for the calculation, but the final continuum expression must be stated in a covariant scalar basis.

Action. The constant-field specialization gives the corresponding pure geometric density,

$$\mathcal{B}_4^{(4)}1(x) = \ell^{-4}\left[\alpha_4 + \beta_4 \sum_{i=1}^{6} C_i N_i(x)\right], \tag{6}$$

where $N_i(x) = |L_i(x)|$. Since $[\mathcal{B}_4^{(4)}] = L^{-4}$, the quantity $\ell^4\mathcal{B}_4^{(4)}1$ is dimensionless. This is precisely the dimension required for a local four-dimensional action contribution. We may therefore define, without introducing any additional dimensional conversion factor,

$$S_4[C] = \lambda_4 \sum_{x\in C} \ell^4 \mathcal{B}_4^{(4)}1(x), \tag{7}$$

where λ_4 is dimensionless.

Equation (7) is the direct analogue, at four derivatives, of the use of the causal-set curvature estimator in the Benincasa-Dowker action. Its defining data are microscopic: the action is a sum over causal-set elements of a function of finite-order interval cardinalities. No continuum metric, coordinate distance, or lattice orientation is required to define the discrete quantity. The metric enters only when one asks for its continuum image.

If the constant-field specialization has the expected local continuum limit, its bulk contribution would have the form

$$\lim_{\rho\to\infty} \langle S_4[C]\rangle = \lambda_4 \int_M d^4x\sqrt{-g}\,\mathcal{Q}_4[g] + S^{(4)}_{\text{bdry}}, \tag{8}$$

where $\mathcal{Q}_4[g]$ would be the curvature-squared combination determined by the scalar reduction of the constant-field sector. Its explicit coefficients have not yet been calculated, so Eq. (8) is the expected continuum organization, not a completed continuum-limit result.

Boundary contributions also require separate analysis: the causal-set action is bilocal at finite sprinkling density, and continuum boundary terms can survive the limit [9, 10, 12]. The boundary completion of the proposed four-derivative contribution therefore remains an open part of the analysis.

If the bulk limit is established, the effective-action organization would be

$$S_{\text{eff}} = S_{\text{EH}} + \lambda_4 \int_M d^4x\sqrt{-g}\,\mathcal{Q}_4[g] + \cdots, \tag{9}$$

where the ellipsis contains higher-derivative terms.

In the conventional continuum description, the four-derivative sector is expressed in a basis of curvature-squared invariants, with relations among them supplied in four dimensions by the Gauss–Bonnet density and total derivatives. The purpose of the causal construction is to select a definite combination from microscopic order data. Establishing that combination requires completing the continuum coefficient calculation.

Why this Matters? The motivation for Eq. (7) is to construct a candidate microscopic contribution to causal-set dynamics whose continuum limit could represent a higher-curvature correction. A causal-set ensemble can now be considered with a hierarchy of action terms, schematically

$$S[C] = S_{\text{BD}}[C] + \lambda_4 S_4[C] + \lambda_6 S_6[C] + \cdots, \tag{10}$$

where each term is built directly from order-theoretic observables of the appropriate engineering dimension. The

four-dimensional S_4 term is the first member beyond the Einstein-Hilbert level whose continuum image has the dimension of a curvature-squared Lagrangian density.

This provides a direct bridge to the broader problem of quantum gravitational dynamics. In asymptotic-safety studies, higher-curvature couplings are possible independent directions of the renormalization group and may be needed to describe the ultraviolet fixed-point structure [2, 3]. A causal-set realization changes the question from how a continuum truncation should be discretized to which combinations of higher-derivative couplings can emerge from causal order itself. The construction of de Brito, Eichhorn, and Pfeiffer already demonstrated that higher-curvature information can be incorporated into a causal-set action through iterated operators [11]; the present operator supplies a direct single-kernel realization of the same derivative order and thereby separates the notion of "higher order" from composition of lower-order maps.

There is also a conceptual lesson concerning locality. Causal-set theory is fundamentally nonlocal because causal intervals contain arbitrarily many elements in the continuum regime, while the local continuum d'Alembertian is a differential operator. The recovery of a local second-order operator is therefore already a nontrivial dynamical fact, and recent work has revisited the compatibility of locality and Lorentz invariance within the causal-set framework [8, 13]. At fourth order, the issue becomes sharper: whether the Poisson mean of the proposed kernel converges to a local $\Box^2$ operator with a covariant curvature completion must be established by the full basis summation. Locality, Lorentz covariance, and the form of the higher-derivative completion are logically distinct properties that must be checked separately.

The distinction from an iterated operator is physically useful. The B^2 construction studied previously gives a particular curvature-squared combination, $R^2 - 2\Box R$, when inserted into the causal-set action [11]. A direct fourth-order kernel is not constrained to inherit the composition rule of B^2 and can instead carry information associated with the full fourth-order causal neighbourhood. This opens the possibility of exploring more than one higher-curvature direction while keeping the microscopic definition linear in the field being acted upon. At the level of the gravitational action, this turns the passage from the Einstein-Hilbert term to higher derivatives into an expansion in distinct causal densities rather than simply powers of one discrete operator.

The boundary structure deserves equal emphasis. In the continuum, curvature-squared densities are related by integrations by parts, Gauss-Bonnet identities, and total derivatives; in a causal set, the finite-range action knows about the causal boundary directly. Existing studies show that the mean causal-set action can acquire boundary and joint terms in the continuum limit [9, 10, 12]. The proposed four-derivative density should therefore be regarded as a candidate bulk contribution to a larger action-level construction.

The dimensional matching has a further practical consequence. Because the fourth-order causal density is already dimensionless after multiplication by ℓ^4, its coefficient can be treated as a genuine dimensionless microscopic coupling, just as a four-dimensional curvature-squared Wilson coefficient is dimensionless in the continuum. This makes the causal-set construction suitable for numerical studies in which one varies the higher-order coupling and measures its effect on the ensemble. It also provides a direct language for asking whether a continuum phase, when it exists, dynamically suppresses or enhances higher-derivative directions.

At the same time, the six-layer result gives a useful notion of microscopic scale separation. The operator probes only finitely many order levels, while the continuum limit is taken at increasing sprinkling density. A mesoscopic version can retain the same continuum expectation while averaging over many fundamental elements, potentially controlling fluctuations without changing the local operator extracted from the mean. This makes the fourth-order action a candidate observable not only for analytic continuum matching but also for numerical studies of causal-set ensembles and their approach to continuum behavior.

Conclusion and Outlook. We have formulated a finite-layer causal-set ansatz intended to represent a fourth-order scalar operator and a corresponding geometric action contribution. In four dimensions, the layer polynomial is supported on six past causal layers. The continuum analysis identifies a finite intermediate basis, but the complete summation and covariant reduction required to determine the final local operator have not yet been completed. We therefore do not claim a fully evaluated curvature completion or an explicit curvature-squared density at this stage.

The remaining calculation is finite but lengthy. It requires summing the intermediate coefficients over p, q, m, showing how the frame-dependent contractions reduce to the Lorentz-invariant scalar basis, fixing β_d by normalizing the coefficient of $\Box^2\phi$ to unity, and fixing α_d by cancellation of the leading density-dependent $\phi(x)$ contribution. The resulting coefficients will determine the complete local continuum operator and the corresponding bulk curvature-squared combination.

We leave this explicit coefficient evaluation for a subsequent version. Until that calculation is completed, the present work should be read as the construction of a candidate finite-layer kernel and the framework for its continuum analysis, rather than as a completed derivation of the full four-derivative continuum operator. The subsequent steps are to establish the explicit continuum coefficients, determine the boundary completion, and investigate fluctuations and the role of the resulting action contribution in causal-set dynamics.

Conventions. We work in a d-dimensional Riemannian/Lorentzian spacetime with a torsion-free connection, meaning the Christoffel symbols are symmetric in their lower indices, $\Gamma^{\mu}_{\nu\rho} = \Gamma^{\mu}_{\rho\nu}$. Our Riemann tensor and covariant derivative commutator conventions are defined by $[\nabla_\mu, \nabla_\nu]V_\rho = R_{\mu\nu\rho}{}^{\sigma}V_\sigma$.

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Appendix A: Continuum-Limit Analysis

We consider the finite-layer causal-set operator

$$\mathcal{B}_4^{(d)}\phi(x) = \frac{1}{\ell^4}\left[\alpha_d\phi(x) + \beta_d \sum_{i=1}^{n_d} C_i^{(d)} \sum_{y\in L_i(x)} \phi(y)\right], \tag{A1}$$

where $L_i(x)$ denotes the ith past layer of x and n_d is the number of layers. We introduce

$$\rho := \ell^{-d}, \qquad \ell^{-4} = \rho^{4/d}, \tag{A2}$$

and define the layer polynomial

$$\mathcal{P}_d(z) := \sum_{i=1}^{n_d} C_i^{(d)} \frac{z^{i-1}}{(i-1)!}. \tag{A3}$$

Poisson averaging then gives

$$\left\langle \mathcal{B}_4^{(d)}\phi(x)\right\rangle = \alpha_d\rho^{4/d}\phi(x) + \beta_d\rho^{1+4/d}\int_{J^-(x)} d^dy\sqrt{-g(y)}\phi(y)\mathcal{P}_d\left(\rho V(x,y)\right)e^{-\rho V(x,y)}, \tag{A4}$$

where $V(x,y)$ is the volume of the Alexandrov interval $A(x,y) = J^+(y)\cap J^-(x)$.

The density-dependent polynomial will be obtained from an Euler operator. We define

$$D_\rho := \rho\frac{\partial}{\partial\rho}. \tag{A5}$$

The construction below determines the minimal polynomial whose action on $e^{-\rho V}$ removes all algebraically non-vanishing contributions from the region down the light cone, thereby generating the layer factor $P_d(\rho V)e^{-\rho V}$.

1. Geometric Setup and Decomposition

Let

$$K := \operatorname{supp}\phi \tag{A6}$$

denote the compact support of the test field. Thus only $K\cap J^-(x)$ contributes to Eq. (A4). We take x to be the origin of a Riemann-normal-coordinate (RNC) neighbourhood $\mathcal{N}$ and introduce the past-directed radial null coordinates

$$u = \frac{-t-r}{\sqrt{2}}, \qquad v = \frac{-t+r}{\sqrt{2}}, \tag{A7}$$

so that locally the past light cone of x is $u = 0$ and the causal past is described by $0 \le u \le v$. The RNC description is convenient near x, but it is not adapted to

the part of the light cone extending away from the origin. We therefore introduce the portion of the past light cone relevant to the support of the field,

$$\mathcal{L} := \partial J^-(x) \cap K, \tag{A8}$$

and assume that each point of $\mathcal{L}$ is connected to x by a unique past-directed null geodesic. The corresponding null generators are denoted by γ_Ω, where

$$\Omega = (\Omega^1, \ldots, \Omega^{d-2}) \tag{A9}$$

labels the generators. In a neighbourhood $\mathcal{N}_{\rm LC}$ of $\mathcal{L}$ we use Null Gaussian Normal Coordinates (NGNC),

$$(U, V, \Omega), \tag{A10}$$

where V is an affine parameter along a null generator and U is an affine parameter along the transverse null geodesics. The light cone itself is

$$U = 0. \tag{A11}$$

The NGNC therefore provide a single coordinate system adapted to the whole null hypersurface. For the local volume estimate around any fixed generator γ_Ω, we further assume the existence of a tubular neighbourhood $\mathcal{T}_\Omega$ of that generator on which Null Fermi Normal Coordinates (NFNC) are valid. These coordinates are adapted to the individual null geodesic and its parallel-transported null frame. We choose the transverse thickness of the light-cone region sufficiently small that the complete Alexandrov interval between x and every point of W_2 remains inside the corresponding $\mathcal{T}_\Omega$. This is the geometric assumption that permits the use of the NFNC expansion and its Penrose-limit scaling. The terminology and construction follow the null Gaussian and null Fermi coordinate treatment used for the curved-spacetime light-cone analysis of Ref. [14], based on the construction of null Fermi coordinates in Ref. [15]. There is no need to force the RNC and NGNC boundaries to coincide. Indeed, they are adapted to different coordinate constructions. In curved spacetime, one generally has

$$u \neq U, \qquad v \neq V \tag{A12}$$

away from the common null hypersurface. The coordinate transition map is nevertheless smooth in the overlap of the two neighbourhoods, and the two descriptions become arbitrarily close as the RNC neighbourhood is shrunk towards x. We therefore choose two small positive scales satisfying

$$0 < a' \leq a, \tag{A13}$$

where a controls the RNC near region and a' is the corresponding lower affine cutoff used in NGNC. The inequality $a' \leq a$ is chosen deliberately: it allows the two regions to overlap rather than forcing an artificial separation between two different coordinate systems. This is the same geometric issue exhibited explicitly in Fig. 2 of Ref. [14], where the RNC and NGNC/NFNC boundaries are shown to be misaligned in curved spacetime.

With these coordinate setups, the near region is

$$W_1 := \{ y \in \mathcal{N} \mid 0 \leq u(y) \leq v(y) \leq a \}. \tag{A14}$$

The light-cone region is defined in NGNC by

$$W_2 := \left\{ y \in \mathcal{N}_{\rm LC} \cap K \cap J^-(x) \;\middle|\; a' \leq V(y) \leq L, \quad 0 \leq U(y) \leq \frac{b^2}{[f_0(V(y), \Omega)]^{2/d}} \right\}, \tag{A15}$$

where L is chosen large enough to contain the relevant support of ϕ along the light cone. The positive constant b has dimensions of length. The f_0-dependent form of the transverse cutoff is useful because it makes the leading Alexandrov volume at the upper boundary of W_2 independent of V and Ω. Indeed, once the light-cone expansion derived below is introduced,

$$V_0 = U^{d/2} f_0(V, \Omega), \tag{A16}$$

and on the upper boundary of W_2, $V_0 \leq b^d$. Finally, define the remaining region by

$$W_3 := (K \cap J^-(x)) \setminus (W_1 \cup W_2). \tag{A17}$$

Because W_1 and W_2 are defined in different coordinate descriptions, and because their boundaries need not coincide in curved spacetime, we do not require a disjoint partition. Instead, the overlap

$$W_{12} := W_1 \cap W_2 \tag{A18}$$

is retained. If

$$\mathscr{I}[W] := \widehat{\mathcal{E}}_d \int_W d^d y \sqrt{-g(y)} \phi(y) e^{-\rho V(x,y)}, \tag{A19}$$

then

$$\mathscr{I}[W_1] + \mathscr{I}[W_2] + \mathscr{I}[W_3] = \mathscr{I}[K \cap J^-(x)] + \mathscr{I}[W_{12}]. \tag{A20}$$

Since

$$W_{12} \subset W_2, \tag{A21}$$

the additional term is included in the light-cone analysis. Once the W_2 contribution is shown to vanish in the continuum limit, the overlap therefore has no effect on the final result. This is the reason it is preferable here not to force a disjoint partition merely for geometrical convenience.

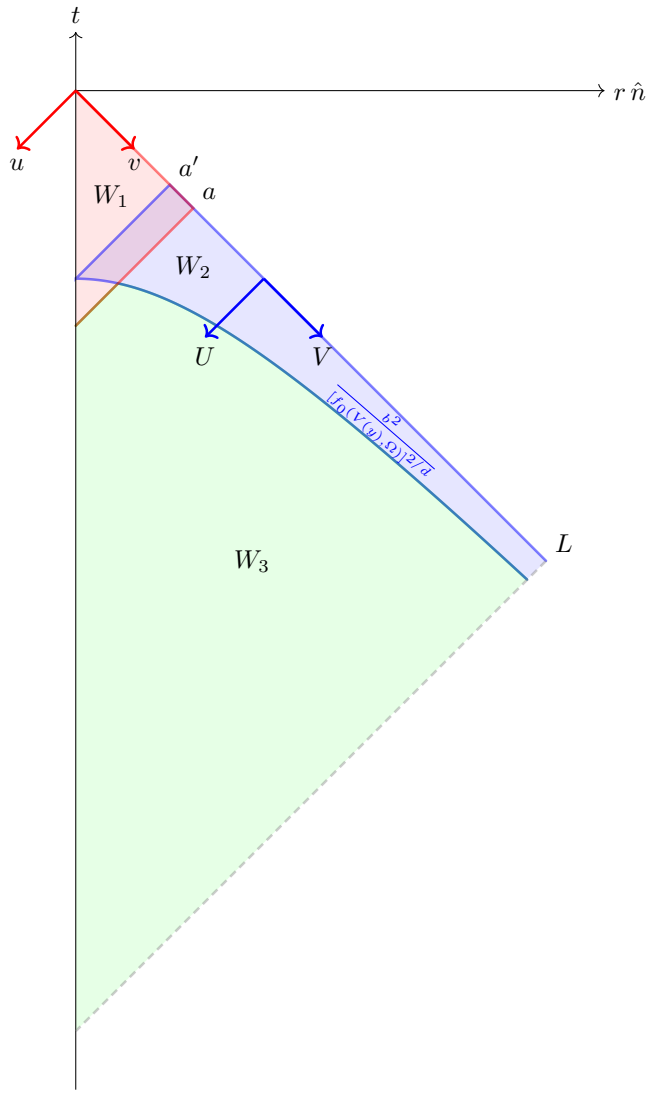


FIG. 1. Schematic geometry of the continuum-limit decomposition. The RNC-defined near region W_1 and the NGNC-defined light-cone region W_2 need not have coincident boundaries in curved spacetime, and we therefore retain their overlap. The transverse thickness of W_2 is chosen using the leading light-cone coefficient $f_0(V,\Omega)$ so that the leading Alexandrov volume remains uniformly bounded throughout the tubular neighbourhood.

2. The Deep-Past Region W_3

By construction, W_3 is bounded away from the past light cone. The Alexandrov volume is continuous in the relevant chronological region and vanishes on the null boundary. Since W_3 is compact and does not intersect that boundary, there exists a finite strictly positive lower bound

$$V(x,y) \geq V_{\min} > 0, \qquad y \in W_3, \tag{A22}$$

where

$$V_{\min} := \min_{y \in W_3} V(x,y). \tag{A23}$$

The precise location of the minimum is immaterial for the continuum limit. In the flat-space construction, the corresponding lower bound can be identified explicitly from the boundary separating the near and deep regions; in particular, the flat geometry provides the familiar reference scale associated with the surface $u = v = a$. In the curved geometry, the boundary is displaced and distorted, so we do not need to identify its minimizing point. What survives from the flat argument is the invariant statement (A22): once a finite neighbourhood of the light cone has been removed, the remaining compact set has a non-zero Alexandrov-volume gap.

For any finite-order density differential operator, differentiating the factor $e^{-\rho V}$ can generate only finite polynomial powers of ρ and factors involving V. Since V is bounded below on W_3, there is therefore a constant C and a finite integer N such that

$$|\mathscr{I}[W_3]| \leq C\rho^N e^{-\rho V_{\min}}. \tag{A24}$$

Hence

$$\lim_{\rho\to\infty} \rho^N e^{-\rho V_{\min}} = 0, \tag{A25}$$

and consequently

$$\lim_{\rho\to\infty} \mathscr{I}[W_3] = 0. \tag{A26}$$

Thus, no local curvature expansion is required in W_3.

3. The Light-Cone Region W_2

The region W_2 requires a different analysis because the Alexandrov volume approaches zero as $U \to 0$. The relevant geometry is controlled by the tubular neighbourhood of the null generators introduced above. For a fixed generator γ_Ω, introduce Null Fermi Normal Coordinates

$$(x^+, x^-, x^a), \qquad a = 1, \ldots, d-2, \tag{A27}$$

with the reference generator represented by

$$x^- = 0, \qquad x^a = 0. \tag{A28}$$

The endpoint y lying on the transverse null geodesic from γ_Ω is represented by

$$(x^+, x^-, x^a) = (V, U, 0, \ldots, 0). \tag{A29}$$

The relevant Penrose scaling is

$$x^+ = z^+, \qquad x^- = Uz^-, \qquad x^a = \sqrt{U} z^a. \tag{A30}$$

The corresponding pulled-back metric has the form

$$ds^2 = U\left[-2dz^+dz^- + \delta_{ab}dz^a dz^b - R_{+a+b}(z^+)z^a z^b (dz^+)^2 + O(U^{1/2})\right], \tag{A31}$$

where the curvature components are evaluated along the reference null generator. Thus, the transverse scaling produces a Jacobian

$$U \times U^{(d-2)/2} = U^{d/2},$$

and the integrated Alexandrov volume has the leading homogeneity

$$V(x,y) = U^{d/2}\left[f_0(V,\Omega) + o(1)\right], \qquad U \to 0^+. \tag{A32}$$

We now define

$$f_0(V,\Omega) := \lim_{U\to 0^+} U^{-d/2} V(x,y). \tag{A33}$$

The function f_0 is finite and positive for $V > 0$. It is determined by the volume of the limiting Alexandrov interval in the corresponding Penrose-limit geometry. Moreover, causal inclusion of the limiting intervals implies

$$0 < V < V' \quad \Longrightarrow \quad f_0(V,\Omega) \le f_0(V',\Omega), \tag{A34}$$

so f_0 is non-decreasing along each null generator. For the asymptotic analysis, it is convenient to retain one controlled subleading order and write

$$V(x,y) = U^{d/2} f_0(V,\Omega) + U^{d/2+1} G(U,V,\Omega), \tag{A35}$$

where G is bounded on the compact W_2 region. This is the general-dimensional form of the regular light-cone volume expansion. The detailed coefficients contained in G will not be required. It is useful to separate

$$V_0 := U^{d/2} f_0(V,\Omega), \qquad \delta V := U^{d/2+1} G(U,V,\Omega), \tag{A36}$$

so that

$$V = V_0 + \delta V. \tag{A37}$$

On the region W_2, the choice of the transverse cutoff in Eq. (A15) gives

$$\frac{\delta V}{V_0} = \frac{UG}{f_0} \tag{A38}$$

and hence

$$\left|\frac{\delta V}{V_0}\right| \le b^2 \frac{\sup_{W_2}|G|}{\inf_{W_2} f_0}\left[\inf_{W_2} f_0\right]^{-2/d}. \tag{A39}$$

Because f_0 is positive and continuous on the compact (V,Ω) domain $[a',L]\times S^{d-2}$, the denominator is strictly positive. The parameter b can therefore be chosen sufficiently small that

$$|\delta V| \ll V_0 \tag{A40}$$

uniformly throughout W_2.

The characteristic transverse scale follows from

$$\rho V_0 \sim 1, \tag{A41}$$

which gives

$$U \sim \rho^{-2/d}. \tag{A42}$$

On this scale,

$$\rho\delta V = \rho U^{d/2+1} G = O(\rho^{-2/d}), \tag{A43}$$

so that the correction to the exponential is subleading:

$$e^{-\rho V} = e^{-\rho V_0}\left[1 + O(\rho^{-2/d})\right]. \tag{A44}$$

Thus the leading density power counting is completely controlled by $U^{d/2} f_0$. To see why the f_0-dependent upper boundary in Eq. (A15) is particularly convenient, consider the basic transverse integral

$$J_m(\rho) := \int_0^{b^2/f_0^{2/d}} dU U^m e^{-\rho f_0 U^{d/2}}. \tag{A45}$$

With

$$s = \rho f_0 U^{d/2}, \tag{A46}$$

we obtain

$$U = \left(\frac{s}{\rho f_0}\right)^{2/d}, \qquad dU = \frac{2}{d}(\rho f_0)^{-2/d} s^{2/d-1} ds, \tag{A47}$$

and therefore

$$J_m(\rho) = \frac{2}{d}(\rho f_0)^{-2(m+1)/d}\gamma\left(\frac{2(m+1)}{d}, \rho b^d\right), \tag{A48}$$

where γ is the lower incomplete Gamma function. Notice that the upper argument of the incomplete Gamma

function is independent of V and Ω. Consequently, for fixed $b > 0$,

$$J_m(\rho) = \frac{2}{d}\Gamma\left(\frac{2(m+1)}{d}\right) f_0(V,\Omega)^{-2(m+1)/d}\rho^{-2(m+1)/d} + \mathcal{O}\left(e^{-\rho b^d}\right). \tag{A49}$$

The measure and the scalar field are smooth in the transverse coordinate, so their product may be expanded locally as

$$\sqrt{-g(y)}\phi(y)d^dy = \left[\sum_{m=0}^{M} A_m(V,\Omega)U^m + O(U^{M+1})\right] \times dU dV d\Omega_{d-2}, \tag{A50}$$

for any finite M required in the asymptotic argument. Combining this expansion with Eq. (A49), the contribution of a term of transverse order U^m to the integral appearing before the Euler operator is

$$\rho^{-2(m+1)/d}. \tag{A51}$$

The overall layer contribution in Eq. (A4) carries the factor $\rho^{1+4/d}$. Thus, before cancellation, a term of order U^m contributes with the density power

$$\rho^{1+4/d-2(m+1)/d} = \rho^{(d+2-2m)/d}. \tag{A52}$$

All terms which do not vanish automatically in the light-cone asymptotic expansion must be annihilated by the density operator. Equivalently, the powers occurring in the transverse integral that need to be removed are

$$\rho^{-2j/d}, \qquad j = 1, \ldots, q_d, \tag{A53}$$

with

$$q_d = \left\lfloor \frac{d+2}{2} \right\rfloor + 1. \tag{A54}$$

Since

$$D_\rho \rho^\lambda = \lambda \rho^\lambda, \tag{A55}$$

the minimal Euler polynomial required by the light-cone region alone is

$$\widehat{\mathcal{E}}_d^{\mathrm{LC}} = \mathcal{N}_d^{\mathrm{LC}} \prod_{j=1}^{q_d}\left(D_\rho + \frac{2j}{d}\right), \tag{A56}$$

where

$$\mathcal{N}_d^{\mathrm{LC}} = \left[\prod_{j=1}^{q_d}\frac{2j}{d}\right]^{-1} = \frac{d^{q_d}}{2^{q_d} q_d!}. \tag{A57}$$

This operator is fixed by the W_2 power counting. The near-region analysis below imposes one additional condition, so $\widehat{\mathcal{E}}_d^{\mathrm{LC}}$ is the light-cone factor of the final density operator rather than the final operator itself. Define

$$\mathscr{I}[W_2] := \widehat{\mathcal{E}}_d^{\mathrm{LC}} \int_{W_2} d^dy \sqrt{-g(y)}\phi(y) e^{-\rho V(x,y)}. \tag{A58}$$

The preceding power counting gives

$$\mathscr{I}[W_2] = O\left(\rho^{-2(q_d+1)/d}\right), \tag{A59}$$

and hence the W_2 contribution vanishes in the continuum limit. The final operator constructed below contains $\widehat{\mathcal{E}}_d^{\mathrm{LC}}$ as a factor, so this conclusion is unchanged by the subsequent W_1 refinement.

4. The Near Region W_1

The near region W_1 contains the point x and is therefore the only region that can contribute a genuinely local term in the continuum limit. Recall that

$$W_1 = \{y \in \mathcal{N} \mid 0 \le u \le v \le a\}, \tag{A60}$$

where $\mathcal{N}$ is the Riemann-normal-coordinate neighbourhood of $x = 0$. Using

$$u = \frac{-y^0 - r}{\sqrt{2}}, \qquad v = \frac{-y^0 + r}{\sqrt{2}}, \qquad r = \frac{v-u}{\sqrt{2}}, \tag{A61}$$

the coordinate volume element becomes

$$d^dy = 2^{-(d-2)/2}(v-u)^{d-2} du dv d\Omega_{d-2}. \tag{A62}$$

We define

$$I_1(\rho) := \widehat{\mathcal{E}}_d \int_{W_1} d^dy \sqrt{-g(y)}\phi(y) e^{-\rho V(x,y)}. \tag{A63}$$

The corresponding contribution to the expectation value in Eq. (A4) is

$$\mathcal{B}^{(d)}_{4,W_1}\phi(x) = \beta_d \rho^{1+4/d} I_1(\rho). \tag{A64}$$

a. Local Expansions and Truncation. The relevant large-density behaviour in W_1 is governed by the flat Alexandrov volume. From Appendix C,

$$V_0(y) := V^{(d)}(y) = \frac{\Omega_{d-2}}{2^{d-1}d(d-1)}\tau^d, \qquad \tau^2 = 2uv. \tag{A65}$$

Thus, ρV_0 is of order unity in the region that controls the large-ρ asymptotics. We assign the near-region weights

$$\mathrm{ord}_{\mathrm{nr}}(y^\mu) = 1, \qquad \mathrm{ord}_{\mathrm{nr}}(\rho) = -d, \qquad \mathrm{ord}_{\mathrm{nr}}(\rho V_0) = 0. \tag{A66}$$

Since V_0 has coordinate order d, a relative correction of coordinate order k to the Alexandrov volume produces

$$\mathrm{ord}_{\mathrm{nr}}\left(\rho \delta V_{(k)}\right) = k.$$

Because the overall layer contribution in Eq. (A64) carries $\rho^{1+4/d}$, terms of effective order four are the last terms that can contribute to a finite continuum limit. We therefore retain all terms through effective order four.

The scalar field is expanded through fourth order,

$$\phi(y) = \phi + \phi_{,\mu} y^\mu + \frac{1}{2}\phi_{,\mu\nu} y^\mu y^\nu + \frac{1}{3!}\phi_{,\mu\nu\rho} y^\mu y^\nu y^\rho + \frac{1}{4!}\phi_{,\mu\nu\rho\sigma} y^\mu y^\nu y^\rho y^\sigma + \mathcal{R}_\phi^{(5)}(y), \tag{A67}$$

where all derivatives are evaluated at $x = 0$, and

$$\mathcal{R}_\phi^{(5)}(y) = y^\mu y^\nu y^\rho y^\sigma y^\lambda \Psi_{\mu\nu\rho\sigma\lambda}(y). \tag{A68}$$

The function $\Psi_{\mu\nu\rho\sigma\lambda}$ is bounded on the compact region W_1. For the invariant volume density, we use the expansion already given in Appendix C,

$$\sqrt{-g(y)} = M_{\le 4}(y) + \mathcal{R}_g^{(5)}(y), \tag{A69}$$

where

$$M_{\le 4}(y) = 1 + M_{(2)}(y) + M_{(3)}(y) + M_{(4)}(y) \tag{A70}$$

contains the explicitly displayed terms in Eq. (C6), while

$$\mathcal{R}_g^{(5)}(y) = y^\mu y^\nu y^\rho y^\sigma y^\lambda T_{\mu\nu\rho\sigma\lambda}(y), \tag{A71}$$

with $T_{\mu\nu\rho\sigma\lambda}$ bounded on W_1. For the Alexandrov volume we use Eq. (C52),

$$V(x,y) = V_0(y)\left[1 + \mathcal{V}_{(2)}(y) + \mathcal{V}_{(3)}(y) + \mathcal{V}_{(4)}(y) + \mathcal{O}(y^5)\right]. \tag{A72}$$

Equivalently, defining

$$\delta V_{(k)}(y) := V_0(y)\mathcal{V}_{(k)}(y), \qquad k = 2,3,4, \tag{A73}$$

we have

$$V(x,y) = V_0(y) + \delta V_{(2)}(y) + \delta V_{(3)}(y) + \delta V_{(4)}(y) + \mathcal{R}_V^{(5)}(y), \tag{A74}$$

where

$$\mathcal{R}_V^{(5)}(y) = V_0(y) O_{\rm nr}(5). \tag{A75}$$

The explicit expressions for $\mathcal{V}_{(2)}$, $\mathcal{V}_{(3)}$ and $\mathcal{V}_{(4)}$ are given in Eqs. (C53)-(C54).

It is convenient to separate the flat exponential,

$$e^{-\rho V(x,y)} = e^{-\rho V_0(y)} e^{-\rho \delta V(y)}, \qquad \delta V := V - V_0. \tag{A76}$$

Through effective order four,

$$e^{-\rho\delta V} = 1 - \rho\delta V_{(2)} - \rho\delta V_{(3)} - \rho\delta V_{(4)} + \frac{1}{2}\rho^2 \delta V_{(2)}^2 + O_{\rm nr}(5). \tag{A77}$$

The quadratic term $\frac{1}{2}\rho^2\delta V_{(2)}^2$ must be retained because it is itself of effective order four. The first omitted terms have effective order five, starting with $\rho^2 \delta V_{(2)}\delta V_{(3)}$.

b. Decomposition of the Integrand. Define

$$\phi_{\le 4}(y) = \phi + \phi_{,\mu} y^\mu + \frac{1}{2}\phi_{,\mu\nu} y^\mu y^\nu + \frac{1}{3!}\phi_{,\mu\nu\rho} y^\mu y^\nu y^\rho + \frac{1}{4!}\phi_{,\mu\nu\rho\sigma} y^\mu y^\nu y^\rho y^\sigma, \tag{A78}$$

and

$$\mathscr{E}_{\le 4} = 1 - \rho\delta V_{(2)} - \rho\delta V_{(3)} - \rho\delta V_{(4)} + \frac{1}{2}\rho^2\delta V_{(2)}^2. \tag{A79}$$

The integrand may then be organized as

$$\sqrt{-g(y)}\phi(y) e^{-\rho V(x,y)} = \big[A(y) + B(y) + C(y)\big] e^{-\rho V_0(y)}, \tag{A80}$$

where A contains all explicitly known local terms through effective order four,

$$A(y) := \Big[M_{\le 4}(y)\phi_{\le 4}(y)\mathscr{E}_{\le 4}(y)\Big]_{\le 4}, \tag{A81}$$

with $[\cdots]_{\le 4}$ denoting truncation to effective order at most four. The group B contains every term involving at least one of the remainders

$$\mathcal{R}_\phi^{(5)}, \qquad \mathcal{R}_g^{(5)}, \qquad \mathcal{R}_V^{(5)}. \tag{A82}$$

Finally, C contains the terms generated by the part of the exponential expansion beyond effective order four. Thus the three groups are distinguished by their mathematical role: A is the explicit local sector to be evaluated, B contains the Taylor and volume remainders, and C is the exponential remainder.

c. Vanishing of B. Every contribution to B contains at least five effective powers of the near-region coordinates. To estimate its large-ρ behaviour, consider a generic monomial of coordinate degree $m + n$. After expansion of the factor $(v - u)^{d-2}$, its contribution is reduced to the basic double-integral family of Appendix D 2. The relevant asymptotic sector scales as

$$\rho^{-(m+n+2)/d}. \tag{A83}$$

For an effective-order-s term,

$$m + n = d - 2 + s, \tag{A84}$$

so that, after restoring the overall factor $\rho^{1+4/d}$,

$$\rho^{1+4/d}\rho^{-(m+n+2)/d} = \rho^{(4-s)/d}. \tag{A85}$$

Since $s \ge 5$ for B, this contribution is suppressed by at least one negative power of ρ. The alternative asymptotic sector gives powers $\rho^{-2(m+1)/d}$. Whenever $m + 1 \le q_d$, these are annihilated by the Euler operator $\widehat{\mathcal{E}}_d$ defined in Eq. (A101). For $m + 1 > q_d$, the resulting density power is already negative after the overall factor in Eq. (A64) is restored. Possible logarithmic factors in the diagonal case do not alter the vanishing conclusion. Therefore,

$$\rho^{1+4/d} I_B(\rho) \longrightarrow 0, \qquad \rho \to \infty. \tag{A86}$$

d. Vanishing of C. The group C begins with the first terms omitted from the exponential expansion in Eq. (A77). Its first possible contribution is

$$\rho^2 \delta V_{(2)} \delta V_{(3)}, \tag{A87}$$

which has effective order five. All subsequent terms have still higher effective order. Hence

$$\mathrm{ord}_{\mathrm{nr}}\, C \geq 5. \tag{A88}$$

The same asymptotic estimate used for B then gives

$$\rho^{1+4/d} I_C(\rho) \longrightarrow 0, \qquad \rho \to \infty. \tag{A89}$$

Thus the continuum-limit calculation in the near region reduces to the explicit local sector A alone.

e. Asymptotic Sector Classification. The explicit local sector in Eq. (A81) is most efficiently analyzed by separating the effective coordinate order from the powers of the zeroth-order exponential variable. Write

$$A(y) = \sum_{p=0}^{4} A_{(p)}(u,v), \tag{A90}$$

where p denotes the total explicit coordinate order. With

$$Z_0 := \rho V_0 = \kappa_d \rho (uv)^{d/2}, \qquad \kappa_d = \frac{\Omega_{d-2}}{2^{d/2-1} d(d-1)}, \tag{A91}$$

a typical term in a fixed Z_0^q sector may be written as

$$Z_0^q u^c v^e, \qquad c+e=p. \tag{A92}$$

After multiplication by the radial measure $(v-u)^{d-2}$, one obtains monomials

$$Z_0^q (v-u)^{d-2} u^c v^e \propto \rho^q \sum_{k=0}^{d-2} (-1)^k \binom{d-2}{k} u^M v^N, \tag{A93}$$

where

$$M = k + c + \frac{qd}{2}, \qquad N = d-2-k+e+\frac{qd}{2}. \tag{A94}$$

Hence

$$M-N = 2(k+c)-(d-2+p), \qquad M+N = d-2+p+qd. \tag{A95}$$

In particular, the condition $M=N$ is independent of the Z_0 sector.

Using the asymptotic decomposition of the double integrals in Appendix D 2, the two sectors scale as

$$Z_0^q \mathcal{I}^{(1)}_{M,N} \sim \rho^{-2(k+c+1)/d}, \tag{A96}$$

$$Z_0^q \mathcal{I}^{(2)}_{M,N} \sim \rho^{-(d+p)/d}. \tag{A97}$$

The first sector is already controlled by the light-cone Euler roots: all non-decaying powers are among $\rho^{-2j/d}$ with $j \leq q_d$, while terms with larger j are already suppressed after the overall factor $\rho^{1+4/d}$ is restored. It remains to classify the second asymptotic sector. After angular averaging, the coefficient of each fixed (p,q) sector is symmetric under

$$A_{(p),q}(u,v) = A_{(p),q}(v,u). \tag{A98}$$

The measure factor has parity

$$(v-u)^{d-2} \longrightarrow (-1)^{d-2}(v-u)^{d-2} \qquad (u \leftrightarrow v). \tag{A99}$$

For even d, the full integrand is symmetric. Since the second asymptotic sector satisfies

$$\mathcal{I}^{(2)}_{M,N} = -\mathcal{I}^{(2)}_{N,M}, \qquad M \neq N, \tag{A100}$$

all off-diagonal contributions cancel pairwise. For odd d, the full integrand is antisymmetric. Consequently, all diagonal coefficients vanish, while the off-diagonal terms remain subject only to the density scaling in Eq. (A97). The resulting classification through $p=4$ is summarized in Table I.

f. Final Density Operator. The W_1 classification shows that one additional Euler factor is required to eliminate the $p=2$ sector. The final density operator is therefore

$$\widehat{\mathcal{E}}_d = \mathcal{N}_d \left(D_\rho + \frac{d+2}{d} \right) \prod_{j=1}^{q_d} \left(D_\rho + \frac{2j}{d} \right). \tag{A101}$$

The normalization for the unit constant term is

$$\mathcal{N}_d = \frac{d^{q_d+1}}{2^{q_d} q_d! (d+2)}. \tag{A102}$$

The role of the additional factor depends on the parity of d. For odd d, the $p=2$ power

$$\rho^{-(d+2)/d} \tag{A103}$$

is not among the light-cone roots, so the new factor supplies the missing simple root. For even d, the same root is already present once in the light-cone operator, but the diagonal integral has the logarithmic form

$$\rho^{-(d+2)/d} \left(A \ln \rho + B \right). \tag{A104}$$

The additional factor therefore makes this root double and removes both the logarithmic and non-logarithmic pieces. The extra factor does not remove the sectors required for the continuum limit. For

$$\lambda_0 = -1, \qquad \lambda_4 = -\frac{d+4}{d}, \tag{A105}$$

one has

$$\lambda_0 + \frac{d+2}{d} = \frac{2}{d} \neq 0, \qquad \lambda_4 + \frac{d+2}{d} = -\frac{2}{d} \neq 0. \tag{A106}$$

	d even		d odd	
p	$M \neq N$	$M = N$	$M \neq N$	$M = N$
0	pair cancellation	leading term, fixes α_d	leading term, fixes α_d	no diagonal
1	pair cancellation	no diagonal	old Euler root	antisymmetry $\Rightarrow 0$
2	pair cancellation	diagonal/log sector	missing Euler root	no diagonal
3	pair cancellation	no diagonal	old Euler root	antisymmetry $\Rightarrow 0$
4	pair cancellation	finite local term	finite off-diagonal term	no diagonal

TABLE I. Classification of the second asymptotic sector in W_1. For even d, off-diagonal terms cancel pairwise, while for odd d diagonal coefficients vanish by antisymmetry of the measure. The $p = 1$ and $p = 3$ sectors are otherwise already removed by the light-cone Euler roots. The only additional unwanted sector is $p = 2$: it is diagonal and logarithmic for even d, and off-diagonal for odd d. The $p = 0$ sector supplies the leading normalization term, while $p = 4$ supplies the finite local contribution.

Thus the additional factor removes $p = 2$ without removing either the leading $p = 0$ contribution or the finite $p = 4$ contribution.

Since the final operator has degree $q_d + 1$ in D_ρ, the number of layers is

$$n_d = q_d + 2 = \left\lfloor \frac{d+2}{2} \right\rfloor + 3. \tag{A107}$$

The corresponding layer polynomial is defined by

$$\widehat{\mathcal{E}}_d e^{-z} = \mathcal{P}_d(z) e^{-z}, \qquad z = \rho V, \tag{A108}$$

with

$$\mathcal{P}_d(z) = \sum_{i=1}^{n_d} C_i^{(d)} \frac{z^{i-1}}{(i-1)!}. \tag{A109}$$

g. Sectorwise Evaluation of the Local Sector. The explicit local contribution is organized as

$$A(y) = \sum_{p=0}^{4} A_{(p)}(u, v). \tag{A110}$$

As established above, the sectors $p = 1, 2, 3$ do not contribute in the continuum limit. We therefore retain $p = 0, 4$. With

$$d^d y = 2^{-(d-2)/2} (v-u)^{d-2} du dv d\Omega_{d-2}, \tag{A111}$$

and

$$Z_0 = \sigma_d (uv)^{d/2}, \qquad \sigma_d = \kappa_d \rho, \qquad \kappa_d = \frac{\Omega_{d-2}}{2^{d/2-1} d(d-1)}, \tag{A112}$$

we use the normalized angular average

$$\langle F \rangle_\Omega := \frac{1}{\Omega_{d-2}} \int_{S^{d-2}} d\Omega_{d-2} F. \tag{A113}$$

The quantities $A_{(p),q}$ and $K_m^{(p,q)}$ below are understood as angular averages in this sense. The angular measure is restored through

$$\int_{S^{d-2}} d\Omega_{d-2} F = \Omega_{d-2} \langle F \rangle_\Omega. \tag{A114}$$

For fixed p and $q = 0, 1, 2$, write

$$A_{(p),q}(u, v) = \sum_{m=0}^{p} K_m^{(p,q)} u^{m + \frac{qd}{2}} v^{p-m+\frac{qd}{2}}. \tag{A115}$$

Multiplying by the radial measure gives

$$(v-u)^{d-2} A_{(p),q}(u, v) = \sum_{r=0}^{d-2+p} L_r^{(p,q)} u^{r+\frac{qd}{2}} v^{d-2+p-r+\frac{qd}{2}}, \tag{A116}$$

where

$$L_r^{(p,q)} = \sum_{m=0}^{p} (-1)^{r-m} \binom{d-2}{r-m} K_m^{(p,q)}, \tag{A117}$$

with the binomial coefficient understood to vanish for $r - m \notin \{0, \ldots, d-2\}$.

Using the double-integral identity of Appendix D, the non-diagonal terms separate into the two asymptotic sectors

$$\mathcal{I}_r^{(p,q,1)} \sim \frac{2 L_r^{(p,q)} a^{d-2+p-2r}}{d(d-2+p-2r)} \Gamma\left(q + \frac{2(r+1)}{d}\right) \sigma_d^{-2(r+1)/d}, \tag{A118}$$

$$\mathcal{I}_r^{(p,q,2)} \sim -\frac{2 L_r^{(p,q)}}{d(d-2+p-2r)} \Gamma\left(q + \frac{d+p}{d}\right) \sigma_d^{-(d+p)/d}, \tag{A119}$$

for $r \neq r_\star^{(p)}$, where

$$r_\star^{(p)} = \frac{d-2+p}{2}. \tag{A120}$$

The first sector scales as

$$\rho^{1+4/d} \mathcal{I}_r^{(p,q,1)} \sim \rho^{(d+2-2r)/d}. \tag{A121}$$

Whenever this power is non-decaying, the exponent $-2(r+1)/d$ is one of the Euler roots of $\widehat{\mathcal{E}}_d$; the remaining values of r are already suppressed. Thus the first sector is removed for both even and odd d. The second sector scales as

$$\rho^{1+4/d} \mathcal{I}_r^{(p,q,2)} \sim \rho^{(4-p)/d}. \tag{A122}$$

Thus $p=0$ produces the leading $\rho^{4/d}$ scaling, while $p=4$ produces a finite scaling. For even d, the off-diagonal terms cancel pairwise, leaving only the diagonal logarithmic sector; for odd d, there is no diagonal term for $p=0,4$, and the off-diagonal second sector remains. The diagonal condition is

$$M=N \quad \Longleftrightarrow \quad r=r_\star^{(p)}=\frac{d-2+p}{2}, \tag{A123}$$

which is possible for $p=0,4$ only when d is even. Hence the logarithmic sector is absent for odd d. For even d, the diagonal contribution has the asymptotic form

$$\begin{aligned}\mathcal{I}_{r_\star}^{(p,q)} \sim & \frac{2L_{r_\star}^{(p,q)}}{d^2}\Gamma\left(q+\frac{d+p}{d}\right)\sigma_d^{-(d+p)/d} \\ & \times\left[\ln(\sigma_d a^d)-\psi\left(q+\frac{d+p}{d}\right)\right].\end{aligned} \tag{A124}$$

Thus the logarithmic sector scales as

$$\mathcal{I}_{r_\star}^{(p,q)} \sim \rho^{-(d+p)/d}\ln\rho. \tag{A125}$$

Since $\widehat{\mathcal{E}}_d\rho^\lambda = Q_d(\lambda)\rho^\lambda$ and $Q_d(-(d+p)/d)=0$ for $p=0,4$ and even d,

$$\widehat{\mathcal{E}}_d\left[\rho^{-(d+p)/d}\ln\rho\right]=Q_d'\left(-\frac{d+p}{d}\right)\rho^{-(d+p)/d}. \tag{A126}$$

Consequently, the surviving structure is entirely fixed by the parity of d: odd d gives the second asymptotic sector, while even d gives the logarithmic sector.

Odd-dimensional case. For odd d and $p=0,4$, the integer $d-2+p$ is odd, so there is no diagonal term. The surviving terms occur in pairs

$$r'=d-2+p-r \qquad 0\le r\le\frac{d-3+p}{2}. \tag{A127}$$

Since the angular-averaged polynomial is symmetric under $u\leftrightarrow v$,

$$L_{r'}^{(p,q)}=(-1)^{d-2}L_r^{(p,q)}=-L_r^{(p,q)}, \tag{A128}$$

and therefore

$$\mathcal{I}_r^{(p,q,2)}+\mathcal{I}_{r'}^{(p,q,2)}=-\frac{4L_r^{(p,q)}}{d(d-2+p-2r)}\Gamma\left(q+\frac{d+p}{d}\right)\sigma_d^{-(d+p)/d}. \tag{A129}$$

Substituting the definition of $L_r^{(p,q)}$ and writing $r=m+j$ gives

$$\sum_{r=0}^{d-2+p}\frac{L_r^{(p,q)}}{d-2+p-2r}=\sum_{m=0}^{p}K_m^{(p,q)}\sum_{j=0}^{d-2}\frac{(-1)^j\binom{d-2}{j}}{d-2+p-2m-2j}. \tag{A130}$$

The inner binomial sum is evaluated in closed form using

$$\sum_{j=0}^{n}\frac{(-1)^j\binom{n}{j}}{a-j}=(-1)^n\frac{\Gamma(n+1)\Gamma(a-n)}{\Gamma(a+1)}. \tag{A131}$$

Setting $n=d-2$ and $a=(d-2+p-2m)/2$ gives

$$\sum_{r=0}^{(d-3+p)/2}\frac{L_r^{(p,q)}}{d-2+p-2r}=-\frac{\Gamma(d-1)}{4}\sum_{m=0}^{p}K_m^{(p,q)}\frac{\Gamma\left(\frac{p+2-d-2m}{2}\right)}{\Gamma\left(\frac{d+p-2m}{2}\right)}.$$

Hence, the complete odd-dimensional contribution from the near region is

$$I_1^{\mathrm{odd}}(\rho)\sim\frac{\Omega_{d-2}\Gamma(d-1)}{2^{(d-2)/2}d}\sum_{p\in\{0,4\}}\sum_{q=0}^{2}\Gamma\left(q+\frac{d+p}{d}\right)Q_d'\left(-\frac{d+p}{d}\right)\sigma_d^{-(d+p)/d}\sum_{m=0}^{p}K_m^{(p,q)}\frac{\Gamma\left(\frac{p+2-d-2m}{2}\right)}{\Gamma\left(\frac{d+p-2m}{2}\right)}. \tag{A132}$$

For odd d, $Q_d(-1)\neq 0$ and $Q_d(-(d+4)/d)\neq 0$, so the $p=0$ and $p=4$ pieces scale as ρ^{-1} and $\rho^{-(d+4)/d}$, respectively. Thus

$$\rho^{1+4/d}I_1^{\mathrm{odd}}(\rho) \tag{A133}$$

contains the leading $\rho^{4/d}\phi(x)$ term from $p=0$ and a finite local term from $p=4$.

Even-dimensional case. For even d, the off-diagonal terms cancel because

$$L_{d-2+p-r}^{(p,q)}=L_r^{(p,q)}, \qquad \mathcal{I}_{d-2+p-r}^{(p,q,2)}=-\mathcal{I}_r^{(p,q,2)}. \tag{A134}$$

Only the diagonal term $r_\star^{(p)} = (d-2+p)/2$ remains. Its coefficient can be written without any dimension-dependent summation as

$$L_{r_\star}^{(p,q)} = (-1)^{(d-2+p)/2}\Gamma(d-1)\sum_{m=0}^{p}\frac{(-1)^m K_m^{(p,q)}}{\Gamma\left(\frac{d+p-2m}{2}\right)\Gamma\left(\frac{d-p+2m}{2}\right)}. \tag{A135}$$

Applying the density operator to the logarithmic sector therefore gives

$$\widehat{\mathcal{E}}_d \mathcal{I}_{r_\star}^{(p,q)} \sim \frac{2L_{r_\star}^{(p,q)}}{d^2}\Gamma\left(q+\frac{d+p}{d}\right)Q_d'\left(-\frac{d+p}{d}\right)\sigma_d^{-(d+p)/d}, \tag{A136}$$

and hence the complete even-dimensional near-region contribution is

$$I_1^{\text{even}}(\rho) \sim \frac{\Omega_{d-2}\Gamma(d-1)}{2^{(d-4)/2}d^2}\sum_{p\in\{0,4\}}\sum_{q=0}^{2}(-1)^{(d-2+p)/2}\Gamma\left(q+\frac{d+p}{d}\right)Q_d'\left(-\frac{d+p}{d}\right)\sigma_d^{-(d+p)/d}\sum_{m=0}^{p}\frac{(-1)^m K_m^{(p,q)}}{\Gamma\left(\frac{d+p-2m}{2}\right)\Gamma\left(\frac{d-p+2m}{2}\right)}. \tag{A137}$$

Thus, for even d, the $p=0$ logarithmic sector is converted into a ρ^{-1} term and gives the leading $\rho^{4/d}\phi(x)$ contribution, while the $p=4$ logarithmic sector is converted into a $\rho^{-(d+4)/d}$ term and gives a finite local contribution after multiplication by $\rho^{1+4/d}$.

Intermediate Basis and Covariant Reduction. For the retained sectors $p=0,4$, we expand the coefficients $K_m^{(p,q)}$ in the following intermediate basis of 30 contractions. Once these coefficients are known, every radial coefficient $L_r^{(p,q)}$ is fixed by Eq. (A115), so no separate list of the $L_r^{(p,q)}$ is required. Only $\mathcal{K}_1,\ldots,\mathcal{K}_9$ are manifest spacetime scalars. The remaining contractions, $\mathcal{K}_{10},\ldots,\mathcal{K}_{30}$, involve components relative to the distinguished 0-direction of the adapted frame and are not, individually, Lorentz-invariant scalars.

After the complete summation over p,q,m, the non-scalar contributions must cancel or combine into the scalar basis $\mathcal{K}_1,\ldots,\mathcal{K}_9$, provided the resulting continuum operator is Lorentz invariant. Lorentz invariance alone does not imply that each of the 21 frame-dependent contributions vanishes separately. This reduction must be established by the explicit coefficient summation.

$$\begin{aligned}
&\mathcal{K}_1 = \phi && \mathcal{K}_2 = \Box^2\phi, && \mathcal{K}_3 = \Box R\phi, && \mathcal{K}_4 = R\Box\phi, && \mathcal{K}_5 = R_{\mu\nu}\nabla^\mu\nabla^\nu\phi,\\
&\mathcal{K}_6 = \nabla^\mu R\nabla_\mu\phi, && \mathcal{K}_7 = R^2\phi, && \mathcal{K}_8 = R_{\mu\nu}R^{\mu\nu}\phi, && \mathcal{K}_9 = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\phi, && \mathcal{K}_{10} = \nabla_0\nabla_0\Box\phi,\\
&\mathcal{K}_{11} = \nabla_0\nabla_0 R\phi, && \mathcal{K}_{12} = R\nabla_0\nabla_0\phi, && \mathcal{K}_{13} = R_{\mu0}\nabla_0\nabla^\mu\phi, && \mathcal{K}_{14} = \nabla_0 R_{\mu0}\nabla^\mu\phi, && \mathcal{K}_{15} = \nabla_0 R\nabla_0\phi,\\
&\mathcal{K}_{16} = R_{\mu0}R^\mu{}_0\phi, && \mathcal{K}_{17} = \Box R_{00}\phi, && \mathcal{K}_{18} = R_{00}\Box\phi, && \mathcal{K}_{19} = \nabla^\mu R_{00}\nabla_\mu\phi, && \mathcal{K}_{20} = R_{\mu0\rho0}\nabla^\mu\nabla^\rho\phi,\\
&\mathcal{K}_{21} = \nabla^\mu R_{\mu0\nu0}\nabla^\nu\phi && \mathcal{K}_{22} = RR_{00}\phi, && \mathcal{K}_{23} = R_{\mu\nu\rho0}R^{\mu\nu\rho}{}_0\phi, && \mathcal{K}_{24} = R_{\mu\nu}R^\mu{}_0{}^\nu{}_0\phi, && \mathcal{K}_{25} = \nabla_0\nabla_0\nabla_0\nabla_0\phi,\\
&\mathcal{K}_{26} = \nabla_0\nabla_0 R_{00}\phi, && \mathcal{K}_{27} = R_{00}\nabla_0\nabla_0\nabla_0\phi, && \mathcal{K}_{28} = \nabla_0 R_{00}\nabla_0\phi, && \mathcal{K}_{29} = R_{00}^2\phi, && \mathcal{K}_{30} = R_{\mu0\rho0}R^\mu{}_0{}^\rho{}_0\phi.
\end{aligned}$$

The Cadabra result is written as

$$K_m^{(p,q)} = \sum_{A=1}^{30} c_{mA}^{(p,q)}\mathcal{K}_A. \tag{A138}$$

h. Determination of α_d and β_d. From Eqs. (A4) and (A64),

$$\left\langle B_4^{(d)}\phi(x)\right\rangle = \alpha_d\rho^{4/d}\phi(x) + \beta_d\rho^{1+4/d}I_1(\rho). \tag{A139}$$

Introduce

$$\kappa_d = \frac{\Omega_{d-2}}{2^{d/2-1}d(d-1)}, \qquad \sigma_d = \kappa_d\rho, \tag{A140}$$

so that

$$\rho^{1+4/d}\sigma_d^{-1} = \kappa_d^{-1}\rho^{4/d}, \qquad \rho^{1+4/d}\sigma_d^{-(d+4)/d} = \kappa_d^{-(d+4)/d}. \tag{A141}$$

The normalization coefficients are determined by two conditions on the fully summed continuum expansion. First, β_d is fixed by requiring the coefficient of $\mathcal{K}_2 = \Box^2\phi$ in the finite local limit to be unity. The coefficient of $\Box^2\phi$ must therefore be extracted from the complete p,q,m sum and its reduction to the scalar basis.

Second, α_d is fixed by demanding cancellation of the leading density-dependent contribution proportional to $\rho^{4/d}\phi(x)$. This cancellation removes the divergent flat-

space term before the finite local contribution is identified.

The remaining step is to perform the complete summation over p, q, m and reduce the 30 intermediate contractions to the nine-element scalar basis. The non-scalar parts of $\mathcal{K}_{10}, \ldots, \mathcal{K}_{30}$ must cancel or combine into scalar structures; their individual vanishing is not assumed.

Since this explicit summation and reduction have not yet been completed, we do not give closed expressions for α_d, β_d, or the complete coefficients of the local operator at this stage.

5. Summary

For an integer spacetime dimension $d \geq 2$, the fourth-order causal-set operator is defined by the finite-layer ansatz

$$\mathcal{B}_4^{(d)}\phi(x) = \frac{1}{\ell^4}\left[\alpha_d\phi(x) + \beta_d \sum_{i=1}^{n_d} C_i^{(d)} \sum_{y\in L_i(x)} \phi(y)\right], \quad \rho = \ell^{-d}. \tag{A142}$$

The number of layers is

$$q_d = \left\lfloor \frac{d}{2} \right\rfloor + 2, \qquad n_d = \left\lfloor \frac{d}{2} \right\rfloor + 4. \tag{A143}$$

The layer coefficients are generated by the polynomial

$$\mathcal{P}_d(z) = \sum_{i=1}^{n_d} C_i^{(d)} \frac{z^{i-1}}{(i-1)!}, \qquad \widehat{\mathcal{E}}_d e^{-z} = \mathcal{P}_d(z)e^{-z}, \qquad z = \rho V. \tag{A144}$$

Here

$$\widehat{\mathcal{E}}_d = \mathcal{N}_d \left(D_\rho + \frac{d+2}{d}\right) \prod_{j=1}^{q_d} \left(D_\rho + \frac{2j}{d}\right), \qquad D_\rho = \rho\frac{\partial}{\partial\rho}, \tag{A145}$$

with

$$\mathcal{N}_d = \frac{d^{q_d+1}}{2^{q_d} q_d!(d+2)}. \tag{A146}$$

For a direct computation of the coefficients $C_i^{(d)}$, define

$$A_r^{(d)} = \frac{1}{q_d!}\begin{bmatrix} q_d+1 \\ r+1 \end{bmatrix}\left(\frac{d}{2}\right)^r, \qquad A_{-1}^{(d)} = A_{q_d+1}^{(d)} = 0. \tag{A147}$$

The layer coefficients are then

$$C_i^{(d)} = (-1)^{i-1}(i-1)! \sum_{r=i-1}^{q_d+1} \left[A_r^{(d)} + \frac{d}{d+2}A_{r-1}^{(d)}\right] \begin{Bmatrix} r \\ i-1 \end{Bmatrix}. \tag{A148}$$

The geometric factor entering the normalization is

$$\Omega_{d-2} = \frac{2\pi^{(d-1)/2}}{\Gamma\left(\frac{d-1}{2}\right)}, \qquad \kappa_d = \frac{\Omega_{d-2}}{2^{d/2-1}d(d-1)}. \tag{A149}$$

Normalization and Local Continuum Form. The explicit normalization coefficients α_d and β_d are determined by the complete summation of the intermediate continuum contributions. We do not state closed forms for them before that calculation has been completed. The two normalization conditions are as follows. The leading term proportional to $\rho^{4/d}\phi(x)$ must cancel, fixing α_d relative to β_d. The coefficient of $\mathcal{K}_2 = \Box^2\phi$ in the finite local contribution must then be unity, fixing the overall normalization β_d.

The target covariant form of the continuum operator is

$$\mathcal{D}_4^{(d)}\phi(x) = \sum_{A=1}^{9} \gamma_A^{(d)}\mathcal{K}_A, \qquad \gamma_2^{(d)} = 1, \qquad \gamma_1^{(d)} = 0, \tag{A150}$$

where $\mathcal{K}_2 = \Box^2\phi$ and the remaining elements are the scalar structures defined in the intermediate-basis discussion above. The condition $\gamma_1^{(d)} = 0$ expresses the cancellation of the standalone $\phi(x)$ contribution. The coefficients $\gamma_3^{(d)}, \ldots, \gamma_9^{(d)}$ remain to be determined by the complete summation and covariant reduction.

Thus, the formal scalar-basis expansion consists of $\Box^2\phi$ and the other eight listed basis elements, with the standalone ϕ contribution required to cancel. We do not claim that the other coefficients have been evaluated or that all eight remaining contributions survive in the final limit.

The remaining calculation is finite but lengthy. Its completion is necessary before the explicit continuum operator and the corresponding curvature-squared density can be stated.

Appendix B: Fluctuations

The continuum-limit calculation above concerns the expectation value of the causal-set operator. On an individual Poisson sprinkling, however, $B_4^{(d)}\phi(x)$ is a random variable. A finite continuum expectation value therefore does not by itself imply that a typical causal-set realization approaches the continuum operator. The relevant quantity is the variance

$$\mathrm{Var}\left(B_4^{(d)}\phi(x)\right) = \left\langle \left(B_4^{(d)}\phi(x)\right)^2 \right\rangle - \left\langle B_4^{(d)}\phi(x)\right\rangle^2. \tag{B1}$$

For the sharp operator,

$$B_4^{(d)}\phi(x) = \ell^{-4}\left[\alpha_d\phi(x) + \beta_d \sum_{i=1}^{n_d} C_i^{(d)} \sum_{y\in L_i(x)} \phi(y)\right], \tag{B2}$$

the $\alpha_d\phi(x)$ term is deterministic and therefore does not contribute to the variance. The fluctuating part is a Pois-

son sum multiplied by ℓ^{-4}. Campbell-Mecke scaling gives

$$\mathrm{Var}\left(B_4^{(d)}\phi(x)\right) = \ell^{-8}\mathcal{V}_d(\phi, g) + o(\ell^{-8}), \tag{B3}$$

where $\mathcal{V}_d(\phi, g)$ is dimensionless after the local variables have been rescaled by the sprinkling density. Indeed, the Poisson measure supplies one factor of ρ, while the region controlling the ultraviolet asymptotics has volume of order ρ^{-1}; hence the remaining integral is dimensionless and finite for a smooth compactly supported field and a finite-layer kernel. The characteristic scaling is therefore

$$\sqrt{\mathrm{Var}\left(B_4^{(d)}\phi(x)\right)} \sim \ell^{-4}. \tag{B4}$$

This is the higher-order analogue of the fluctuation problem encountered for the causal-set d'Alembertian. The cancellations which render the expectation value finite act on the mean and do not provide a corresponding suppression of the shot noise. Thus increasing the sprinkling density alone does not produce a vanishing fluctuation amplitude for the sharp fourth-order operator.

A natural way to introduce statistical averaging is to separate the fundamental discreteness scale ℓ from a larger mesoscopic nonlocality scale ℓ_k,

$$\ell \ll \ell_k \ll L_{\mathrm{macro}}, \tag{B5}$$

and to define

$$\rho_k := \ell_k^{-d}, \qquad \epsilon := \frac{\rho_k}{\rho} = \left(\frac{\ell}{\ell_k}\right)^d, \qquad 0 < \epsilon < 1. \tag{B6}$$

The sharp operator is recovered at $\epsilon = 1$. For the fourth-order operator, the analogue of the Poisson-thinning construction is obtained by requiring the generating transform

$$F_\epsilon^{(4)}(\xi) := e^{-\xi}\sum_{n=0}^{\infty} f_d^{(4)}(n,\epsilon)\frac{\xi^n}{n!} = \epsilon^{1+4/d}e^{-\epsilon\xi}P_d(\epsilon\xi). \tag{B7}$$

A convenient realization is

$$f_d^{(4)}(n,\epsilon) = \epsilon^{1+4/d}(1-\epsilon)^n\sum_{i=1}^{n_d} C_i^{(d)}\binom{n}{i-1}\left(\frac{\epsilon}{1-\epsilon}\right)^{i-1}. \tag{B8}$$

Indeed, for a fixed layer i, putting $r = i-1$ gives

$$e^{-\xi}\sum_{n=r}^{\infty}(1-\epsilon)^n\binom{n}{r}\left(\frac{\epsilon}{1-\epsilon}\right)^r\frac{\xi^n}{n!} = e^{-\epsilon\xi}\frac{(\epsilon\xi)^r}{r!}, \tag{B9}$$

and summing over i yields Eq. (B7). The corresponding smeared operator is defined by

$$B_{\epsilon,4}^{(d)}\phi(x) = \rho_k^{4/d}\left[\alpha_d\phi(x) + \beta_d\sum_{y\prec x} f_d^{(4)}\left(n(x,y),\epsilon\right)\phi(y)\right]. \tag{B10}$$

Its expectation value is

$$\left\langle B_{\epsilon,4}^{(d)}\phi(x)\right\rangle = \rho_k^{4/d}\left[\alpha_d\phi(x) + \beta_d\rho\int_{J^-(x)} d^dy\sqrt{-g(y)}\phi(y)F_\epsilon^{(4)}\left(\rho V(x,y)\right)\right]. \tag{B11}$$

Therefore,

$$\left\langle B_{\epsilon,4}^{(d)}\phi(x)\right\rangle = \left\langle B_4^{(d)}\phi(x)\right\rangle\Big|_{\rho\to\rho_k}. \tag{B12}$$

The smearing consequently does not modify the continuum operator extracted from the expectation value; it replaces the microscopic density by the effective density ρ_k.

The fluctuation problem is therefore reduced to determining how the variance scales with the two densities. The operator carries the prefactor $\rho_k^{4/d} = \ell_k^{-4}$, while the smeared kernel averages over a region containing, parametrically,

$$N_k \sim \rho\ell_k^d = \left(\frac{\ell_k}{\ell}\right)^d = \epsilon^{-1} \tag{B13}$$

fundamental sprinkling elements. The resulting Poisson suppression gives

$$\mathrm{Var}\left(B_{\epsilon,4}^{(d)}\phi(x)\right) \sim \epsilon\ell_k^{-8} = \frac{\ell^d}{\ell_k^{d+8}}, \tag{B14}$$

provided the dimensionless rescaled second-moment and connected two-pointkernels remain bounded in the local scaling regime. Consequently,

$$\sqrt{\mathrm{Var}\left(B_{\epsilon,4}^{(d)}\phi(x)\right)} \sim \sqrt{\epsilon}\ell_k^{-4} = \frac{\ell^{d/2}}{\ell_k^{d/2+4}}. \tag{B15}$$

Thus, the mesoscopic scale can suppress the sharp fluctuations. There are, however, two independent requirements in a simultaneous continuum limit:

$$\ell_k \longrightarrow 0, \qquad \frac{\ell}{\ell_k} \longrightarrow 0, \tag{B16}$$

so that the operator remains local on macroscopic scales while averaging over many microscopic sprinkling elements, together with the stronger condition

$$\frac{\ell^d}{\ell_k^{d+8}} \longrightarrow 0. \tag{B17}$$

A sufficient hierarchy is therefore

$$\ell \ll \ell_k \ll L_{\text{macro}}, \qquad \frac{\ell^d}{\ell_k^{d+8}} \longrightarrow 0. \tag{B18}$$

Hence the fourth-order causal-set operator admits a two-scale continuum limit in which its expectation value approaches the local continuum operator while its fluctuations are simultaneously driven to zero.

Appendix C: Alexandrov Volume

For two causally related events $y \prec x$, the Alexandrov interval is the intersection of the causal future of the earlier event with the causal past of the later event,

$$A(x,y) := J^+(y) \cap J^-(x). \tag{C1}$$

Its spacetime volume is

$$V(x,y) = \int_{A(x,y)} d^d z \sqrt{-g(z)}. \tag{C2}$$

Thus the Alexandrov volume depends on both the invariant volume element and the causal boundaries determined by the metric.

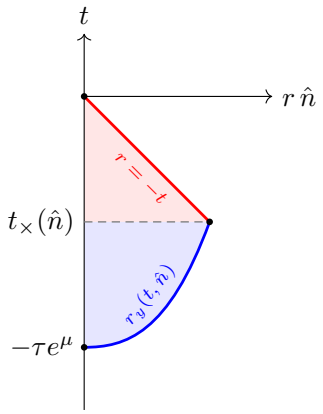


FIG. 2. Schematic Alexandrov interval in the adapted RNC frame. The earlier endpoint lies on the adapted time axis, while the spatial radial direction $\hat{n}$ ranges over the complete $(d-2)$-sphere. In the curved geometry, the future null boundary of y is perturbed, while the past null boundary of the RNC origin remains exactly radial. The shading indicates the two integration regions, separated by the transition time $t_\times(\hat{n})$).

In the present calculation, the later endpoint is placed at the Riemann-normal-coordinate origin, $x^\mu = 0$. Although the separation vector y^μ is not generally aligned with the time coordinate of the original RNC system, it is timelike. We therefore introduce an orthonormal frame adapted to y^μ before determining the integration domain. In this frame the same timelike vector has purely temporal components, which places the endpoint at the centre of the spatial radial coordinates and permits the angular integration to be performed over the full sphere. The invariant proper-time separation is

$$\tau^2 := -\eta_{\mu\nu} y^\mu y^\nu = -2\sigma(0,y) > 0, \tag{C3}$$

where σ is Synge's world function. All curvature tensors appearing below are evaluated at the RNC origin.

1. Riemann Normal Coordinates

The Riemann normal coordinates are constructed about the origin $x^\mu = 0$, such that every geodesic issuing from the origin is represented by a straight coordinate line. At the origin,

$$g_{\mu\nu}(0) = \eta_{\mu\nu}, \qquad \Gamma^\mu{}_{\alpha\beta}(0) = 0, \qquad \partial_\rho g_{\mu\nu}(0) = 0. \tag{C4}$$

Consequently, the metric begins at quadratic order in the coordinates. Since the Alexandrov volume is required through total coordinate order $d+4$, the metric is retained through fourth order,

$$g_{\mu\nu}(x) = \eta_{\mu\nu} - \frac{1}{3}R_{\mu\alpha\nu\beta}(0)x^\alpha x^\beta - \frac{1}{6}\nabla_\gamma R_{\mu\alpha\nu\beta}(0)x^\alpha x^\beta x^\gamma + \frac{1}{180}\left[8R_{\mu\alpha\beta\lambda}(0)R^\lambda{}_{\gamma\delta\nu}(0) - 9\nabla_\gamma\nabla_\delta R_{\mu\alpha\nu\beta}(0)\right]x^\alpha x^\beta x^\gamma x^\delta + \mathcal{O}(x^5). \tag{C5}$$

The corresponding invariant volume density is

$$\sqrt{-g(x)} = 1 - \frac{1}{6}R_{\mu\nu}x^\mu x^\nu - \frac{1}{12}\nabla_\rho R_{\mu\nu}x^\mu x^\nu x^\rho - \left[\frac{1}{40}\nabla_\rho\nabla_\sigma R_{\mu\nu} + \frac{1}{180}R_{\mu\alpha\nu}{}^\lambda R_{\rho\lambda\sigma}{}^\alpha - \frac{1}{72}R_{\mu\nu}R_{\rho\sigma}\right]x^\mu x^\nu x^\rho x^\sigma + \mathcal{O}(x^5). \tag{C6}$$

To make the endpoint geometry compatible with spherical integration, we now choose an orthonormal frame adapted to the timelike separation vector. Define

$$e^\mu := -\frac{y^\mu}{\tau}, \qquad e^\mu e_\mu = -1, \qquad y^\mu = -\tau e^\mu. \tag{C7}$$

Choose $d-1$ spacelike frame vectors $E^\mu{}_{(i)}$ satisfying

$$e_\mu E^\mu{}_{(i)} = 0, \qquad E_{(i)}\cdot E_{(j)} = \delta_{ij}, \qquad i,j = 1,\ldots,d-1. \tag{C8}$$

Together with

$$E^\mu{}_{(0)} := e^\mu, \tag{C9}$$

they form an orthonormal frame at the RNC origin,

$$\eta_{\mu\nu}E^\mu{}_{(A)}E^\nu{}_{(B)} = \eta_{AB}. \tag{C10}$$

The corresponding adapted RNC coordinates are related to the original RNC coordinates by the constant linear transformation

$$x^\mu = E^\mu{}_{(A)}X^{(A)}. \tag{C11}$$

Hence the coordinate volume element is unchanged. In this basis, the endpoint has components

$$y^{(A)} = (-\tau, 0, \ldots, 0). \tag{C12}$$

This is a choice of basis, not a restriction on the original vector y^μ. The invariant statement remains $y^\mu = -\tau e^\mu$. The spatial radial direction is described by a unit vector n^i,

$$n^i n_i = 1, \tag{C13}$$

with covariant spacetime representative

$$\widehat{n}^\mu := E^\mu{}_{(i)}n^i, \qquad e_\mu\widehat{n}^\mu = 0, \qquad \widehat{n}_\mu\widehat{n}^\mu = 1. \tag{C14}$$

The integration point is therefore written as

$$x^\mu = te^\mu + r\widehat{n}^\mu, \tag{C15}$$

with

$$d^dx = dtdrr^{d-2}d\Omega_{d-2}. \tag{C16}$$

The spatial projector associated with the adapted frame is

$$h^{\mu\nu} = \eta^{\mu\nu} + e^\mu e^\nu. \tag{C17}$$

These relations are the only properties of the adapted frame needed below. In particular, the angular integrations are carried out entirely in the Euclidean spatial subspace orthogonal to e^μ and are subsequently reconstructed covariantly.

2. Integration Domain

The causal boundary of the Alexandrov interval consists of the future light cone of y and the past light cone of the origin. Because the Riemann normal coordinates are centred at the origin, every geodesic issuing from the origin is a straight coordinate line. Thus its past null boundary is exact:

$$r_{\rm p}(t) = -t, \qquad t \le 0. \tag{C18}$$

The nontrivial boundary is the future light cone of the endpoint y. It is convenient to determine it from Synge's world function. For nearby points,

$$2\sigma(x,y) = \int_0^1 ds g_{\mu\nu}(z(s))\dot{z}^\mu(s)\dot{z}^\nu(s), \qquad \dot{z}^\mu := \frac{dz^\mu}{ds}, \tag{C19}$$

where $z^\mu(s)$ is the unique geodesic joining the two points, with $z^\mu(0) = y^\mu$, and $z^\mu(1) = x^\mu$. Introducing

$$\Delta^\mu := x^\mu - y^\mu, \tag{C20}$$

we assign coordinate order $\deg x^\mu = \deg y^\mu = 1$; a parenthesized label (k) on a quantities, z, r, t, F and G, denote its homogeneous component of coordinate order k. Thus, the geodesic is expanded perturbatively as

$$z^\mu(s) = z^\mu_{(1)}(s) + z^\mu_{(3)}(s) + \mathcal{O}(\lambda^4), \tag{C21}$$

where

$$z^\mu_{(1)}(s) = y^\mu + s\Delta^\mu, \tag{C22}$$

$$z^\mu_{(3)}(s) = \frac{1}{3}R^\mu{}_{\alpha\beta\nu}\Delta^\alpha\Delta^\beta y^\nu s(s-1). \tag{C23}$$

The higher-order terms are not written explicitly, since only the terms above are required for the world-function expansion used below. The construction satisfies the endpoint conditions order by order.

The null boundary is characterized by

$$\sigma(x, y) = 0. \tag{C24}$$

Using the adapted spherical coordinates (C15),

$$x^\mu = te^\mu + r\widehat{n}^\mu, \qquad \widehat{n}^\mu = E^\mu{}_{(i)} n^i, \qquad y^\mu = -\tau e^\mu, \tag{C25}$$

define

$$F(r; t, n, y) := 2\sigma\left(te + r\widehat{n}, y\right). \tag{C26}$$

Its local expansion takes the form

$$F = F^{(2)} + F^{(4)} + F^{(5)} + F^{(6)} + \mathcal{O}(\lambda^7), \tag{C27}$$

where λ counts coordinate order. In the adapted frame, the leading piece is simply

$$F^{(2)} = r^2 - (t + \tau)^2. \tag{C28}$$

Thus the physicaleading-orderer radial solution is

$$r^{(1)}(t) = t + \tau, \qquad -\tau \le t \le 0. \tag{C29}$$

The algebraic root $r^{(1)} = -(t + \tau)$ is not part of the spherical coordinate domain and is discarded. For sufficiently small timelike separation, the physical curved-space branch remains the perturbative continuation of $r^{(1)}$.

We denote this physical radial boundary by

$$r_y(t, n) = r^{(1)}(t) + \delta r^{(3)}(t, n) + \delta r^{(4)}(t, n) + \delta r^{(5)}(t, n) + \mathcal{O}(\lambda^6). \tag{C30}$$

Substitution into $F(r_y; t, n, y) = 0$ gives the universal perturbative root formulas

$$\delta r^{(3)} = -\left.\frac{F^{(4)}}{\partial_r F^{(2)}}\right|_{r=r^{(1)}}, \tag{C31}$$

$$\delta r^{(4)} = -\left.\frac{F^{(5)}}{\partial_r F^{(2)}}\right|_{r=r^{(1)}}, \tag{C32}$$

$$\delta r^{(5)} = -\left.\frac{F^{(6)} + (\partial_r F^{(4)})\delta r^{(3)} + \frac{1}{2}(\partial_r^2 F^{(2)})(\delta r^{(3)})^2}{\partial_r F^{(2)}}\right|_{r=r^{(1)}}. \tag{C33}$$

All quantities on the right-hand side are evaluated at the corresponding leading-order root.

The radial boundary $r_y(t, n)$ intersects the exact past null boundary of the origin at a single transition time. Define

$$t_\times(n) \quad \text{by} \quad r_y(t_\times, n) = -t_\times. \tag{C34}$$

Equivalently,

$$G(t, n) := F(-t; t, n, y) = 0. \tag{C35}$$

At leading order,

$$G^{(2)}(t) = -\tau^2 - 2\tau t, \tag{C36}$$

so that

$$t_\times^{(1)} = -\frac{\tau}{2}. \tag{C37}$$

We write the curved transition time as

$$t_\times(n) = -\frac{\tau}{2} + \delta t^{(3)}(n) + \delta t^{(4)}(n) + \delta t^{(5)}(n) + \mathcal{O}(\lambda^6). \tag{C38}$$

The corrections are obtained from the same perturbative root construction,

$$\delta t^{(3)} = -\left.\frac{G^{(4)}}{\partial_t G^{(2)}}\right|_{t=-\tau/2}, \tag{C39}$$

$$\delta t^{(4)} = -\left.\frac{G^{(5)}}{\partial_t G^{(2)}}\right|_{t=-\tau/2}, \tag{C40}$$

$$\delta t^{(5)} = -\left.\frac{G^{(6)} + (\partial_t G^{(4)})\delta t^{(3)} + \frac{1}{2}(\partial_t^2 G^{(2)})(\delta t^{(3)})^2}{\partial_t G^{(2)}}\right|_{t=-\tau/2}. \tag{C41}$$

The resulting integration domain is therefore

$$-\tau \le t \le t_\times(n), \qquad 0 \le r \le r_y(t, n), \tag{C42}$$

followed by

$$t_\times(n) \le t \le 0, \qquad 0 \le r \le -t. \tag{C43}$$

The lower radial limit is consequently always $r = 0$ in the adapted-frame calculation. Curvature enters through the deformation of the first upper boundary and through the angular dependence of the transition time.

3. Volume Integral

Using the RNC volume density and the integration domain above, the Alexandrov volume is

$$V(0, y) = \int d\Omega_{d-2}\left[\int_{-\tau}^{t_\times(n)} dt \int_0^{r_y(t,n)} dr r^{d-2}\sqrt{-g(t, r\widehat{n})} + \int_{t_\times(n)}^{0} dt \int_0^{-t} dr r^{d-2}\sqrt{-g(t, r\widehat{n})}\right]. \tag{C44}$$

At this stage, all angular dependence is carried by the spatial unit vector n^i through $\widehat{n}^\mu = E^\mu{}_{(i)} n^i$. It is therefore useful to introduce the spatial projector

$$h^{\mu\nu} = \eta^{\mu\nu} + e^\mu e^\nu, \qquad e_\mu \widehat{n}^\mu = 0, \qquad \widehat{n}_\mu \widehat{n}^\mu = 1. \tag{C45}$$

The angular averages can then be written directly in spacetime form:

$$\int d\Omega_{d-2}\widehat{n}^{\mu}\widehat{n}^{\nu} = \frac{\Omega_{d-2}}{d-1}h^{\mu\nu}, \tag{C46}$$

$$\int d\Omega_{d-2}\widehat{n}^{\mu}\widehat{n}^{\nu}\widehat{n}^{\rho}\widehat{n}^{\sigma} = \frac{\Omega_{d-2}}{(d-1)(d+1)}\Big(h^{\mu\nu}h^{\rho\sigma} + h^{\mu\rho}h^{\nu\sigma} + h^{\mu\sigma}h^{\nu\rho}\Big), \tag{C47}$$

$$\int d\Omega_{d-2}\widehat{n}^{\mu_1}\cdots\widehat{n}^{\mu_{2k+1}} = 0. \tag{C48}$$

Higher even moments are obtained analogously as sums over all pairwise contractions with $h^{\mu\nu}$. The radial integrals are expanded consistently in coordinate order, retaining all terms that contribute through total order $d+4$.

After the angular integrations, the result is covariantized using

$$e^{\mu} = -\frac{y^{\mu}}{\tau}, \qquad h^{\mu\nu} = \eta^{\mu\nu} + \frac{y^{\mu}y^{\nu}}{\tau^2}. \tag{C49}$$

All remaining frame-dependent contractions are thereby expressed in terms of the invariant separation τ and the endpoint vector y^{μ}. The flat-space contribution is

$$V^{(d)}(y) = \frac{\Omega_{d-2}}{2^{d-1}d(d-1)}\tau^d, \qquad \Omega_{d-2} = \frac{2\pi^{(d-1)/2}}{\Gamma\left(\frac{d-1}{2}\right)}. \tag{C50}$$

There is no contribution at order $d+1$ and $d+3$. The local volume expansion required for the present calculation is therefore

$$V(0,y) = V^{(d)}(y) + V^{(d+2)}(y) + V^{(d+4)}(y) + \mathcal{O}(\lambda^{d+5}). \tag{C51}$$

It is convenient to collect the result in the form

$$V(0,y) = \frac{\Omega_{d-2}}{2^{d-1}d(d-1)}\tau^d\left[1 + \mathcal{V}_{(2)} + \mathcal{V}_{(4)} + \mathcal{O}(\lambda^5)\right], \tag{C52}$$

where

$$\mathcal{V}_{(2)} = \frac{d}{24(d+1)}\left[R_{\mu\nu}y^{\mu}y^{\nu} - \frac{R}{d+2}\tau^2\right], \tag{C53}$$

$$\begin{aligned}
\mathcal{V}_{(4)} = {} & \frac{d}{1152(d+1)(d+3)(d+4)}R^2\tau^4 \\
& + \frac{d}{720(d+1)(d+3)(d+4)}R_{\mu\nu}R^{\mu\nu}\tau^4 \\
& - \frac{d}{960(d+1)(d+3)(d+4)}R_{\alpha\beta\gamma\kappa}R^{\alpha\gamma\beta\kappa}\tau^4 \\
& - \frac{d}{576(d+1)(d+3)}RR_{\mu\nu}y^{\mu}y^{\nu}\tau^2 \\
& + \frac{d(7d+26)}{1440(d+1)(d+2)(d+3)}R_{\mu\nu}R^{\mu}{}_{\sigma}y^{\nu}y^{\sigma}\tau^2 \\
& - \frac{d(11d+34)}{1440(d+1)(d+2)(d+3)}R_{\mu\nu}R^{\mu}{}_{\sigma}{}^{\nu}{}_{\beta}y^{\beta}y^{\sigma}\tau^2 \\
& + \frac{d}{480(d+1)(d+3)}R_{\alpha\beta\gamma\kappa}R^{\alpha\gamma\beta}{}_{\sigma}y^{\kappa}y^{\sigma}\tau^2 \\
& + \frac{d(d+2)}{1152(d+1)(d+3)}R_{\mu\nu}R_{\rho\sigma}y^{\mu}y^{\nu}y^{\rho}y^{\sigma} \\
& + \frac{d(7d+22)}{2880(d+1)(d+3)}R_{\alpha\beta\gamma\kappa}R^{\alpha}{}_{\nu}{}^{\gamma}{}_{\sigma}y^{\beta}y^{\kappa}y^{\nu}y^{\sigma}.
\end{aligned} \tag{C54}$$

All curvature tensors and their covariant derivatives in Eqs. (C53)-(C54) are evaluated at the RNC origin. The result is expressed entirely in terms of the invariant proper-time separation and covariant contractions with the endpoint vector y^{μ}.

Appendix D: Auxiliary Identities

1. Covariantization of the Scalar Taylor Expansion

All scalar-field derivatives below are evaluated at the RNC origin x. At this point,

$$g_{\mu\nu}(x) = \eta_{\mu\nu}, \qquad \partial_{\rho}g_{\mu\nu}(x) = 0, \qquad \Gamma^{\rho}_{\mu\nu}(x) = 0 \tag{D1}$$

Moreover, along the radial geodesic generated by a tangent vector y^{μ}, the tangent is parallel transported. Hence

$$\frac{d^n}{ds^n}\phi\left(\exp_x(sy)\right)\Big|_{s=0} = y^{\mu_1}\cdots y^{\mu_n}\nabla_{(\mu_1}\cdots\nabla_{\mu_n)}\phi(x). \tag{D2}$$

Since the ordinary partial derivatives of a scalar commute, comparison with the RNC Taylor expansion gives

$$\partial_\mu \phi = \nabla_\mu \phi \tag{D3}$$
$$\partial_\mu \partial_\nu \phi = \nabla_{(\mu} \nabla_{\nu)} \phi = \nabla_\mu \nabla_\nu \phi \tag{D4}$$
$$\partial_\mu \partial_\nu \partial_\rho \phi = \nabla_{(\mu} \nabla_\nu \nabla_{\rho)} \phi \tag{D5}$$
$$\partial_\mu \partial_\nu \partial_\rho \partial_\sigma \phi = \nabla_{(\mu} \nabla_\nu \nabla_\rho \nabla_{\sigma)} \phi \tag{D6}$$

More generally, the unsymmetrized fourth-order covariant derivative evaluated at the RNC origin is given by

$$\begin{aligned}\nabla_\mu \nabla_\nu \nabla_\rho \nabla_\sigma \phi = {} & \partial_\mu \partial_\nu \partial_\rho \partial_\sigma \phi \\ & + \frac{1}{3}\left(\nabla_\mu R^\lambda{}_{\rho\sigma\nu} + \nabla_\mu R^\lambda{}_{\sigma\rho\nu}\right) \nabla_\lambda \phi \\ & + \frac{1}{3}\left(R^\lambda{}_{\rho\sigma\nu} + R^\lambda{}_{\sigma\rho\nu}\right) \nabla_{\lambda\mu} \phi \\ & + \frac{1}{3}\left(R^\lambda{}_{\rho\sigma\mu} + R^\lambda{}_{\sigma\rho\mu}\right) \nabla_{\lambda\nu} \phi \\ & + \frac{1}{3}\left(R^\lambda{}_{\nu\sigma\mu} + R^\lambda{}_{\sigma\nu\mu}\right) \nabla_{\lambda\rho} \phi \\ & + \frac{1}{3}\left(R^\lambda{}_{\nu\rho\mu} + R^\lambda{}_{\rho\nu\mu}\right) \nabla_{\lambda\sigma} \phi. \end{aligned} \tag{D7}$$

The fourth-order contraction requires the commutation of covariant derivatives. With

$$\Box \phi := g^{\mu\nu} \nabla_\mu \nabla_\nu \phi \tag{D8}$$

and the Riemann-tensor convention used throughout the calculation, one finds

$$g^{\mu\rho} g^{\nu\sigma} \nabla_\mu \nabla_\nu \nabla_\rho \nabla_\sigma \phi = \Box^2 \phi + R^{\mu\nu} \nabla_\mu \nabla_\nu \phi + \frac{1}{2} \nabla^\mu R \nabla_\mu \phi \tag{D9}$$

The remaining non-equivalent contraction gives the same result, and therefore the fully symmetrized fourth-order contraction is

$$g^{\mu\nu} g^{\rho\sigma} \nabla_{(\mu} \nabla_\rho \nabla_\nu \nabla_{\sigma)} \phi = \Box^2 \phi + \frac{2}{3} R^{\mu\nu} \nabla_\mu \nabla_\nu \phi + \frac{1}{3} \nabla^\mu R \nabla_\mu \phi \tag{D10}$$

2. Integral Identities

We use the Gamma function

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt, \tag{D11}$$

together with the digamma function

$$\psi(z) := \frac{d}{dz} \ln \Gamma(z) = \frac{\Gamma'(z)}{\Gamma(z)}. \tag{D12}$$

For later use, we write the lower and upper incomplete Gamma functions as $\gamma(s,z)$ and $\Gamma(s,z)$, respectively, so that

$$\gamma(s,z) + \Gamma(s,z) = \Gamma(s). \tag{D13}$$

a. Exponential Integral. For $n > -1$,

$$\int_0^R x^n e^{-x} dx = \gamma(n+1, R) = \Gamma(n+1) - \Gamma(n+1, R). \tag{D14}$$

For integer $n \geq 0$, this may also be written as

$$\int_0^R x^n e^{-x} dx = n! \left[1 - e^{-R} \sum_{k=0}^{n} \frac{R^k}{k!}\right]. \tag{D15}$$

For fixed n and $R \to \infty$,

$$\Gamma(n+1, R) = R^n e^{-R} \left[1 + \frac{n}{R} + \mathcal{O}\left(R^{-2}\right)\right], \tag{D16}$$

and therefore

$$\int_0^R x^n e^{-x} dx = \Gamma(n+1) - R^n e^{-R} \left[1 + \frac{n}{R} + \mathcal{O}\left(R^{-2}\right)\right]. \tag{D17}$$

b. General-Dimensional Double Integral. Consider

$$\mathcal{I}^{(d)}_{m,n}(a,\sigma) := \int_0^a dv \int_0^v du u^m v^n \exp\left[-\sigma (uv)^{d/2}\right]. \tag{D18}$$

We introduce

$$t = uv, \qquad u = \frac{t}{v}, \qquad du = \frac{dt}{v}. \tag{D19}$$

For fixed v, the original bound $0 \leq u \leq v$ gives $0 \leq t \leq v^2$. Therefore,

$$\mathcal{I}^{(d)}_{m,n} = \int_0^a dv \int_0^{v^2} dt t^m v^{n-m-1} \exp\left[-\sigma t^{d/2}\right]. \tag{D20}$$

The transformed integration region is therefore $0 \leq v \leq a$ and $0 \leq t \leq v^2$. Reversing the order of integration, t runs from 0 to a^2, while for fixed t one has $\sqrt{t} \leq v \leq a$. Hence

$$\mathcal{I}^{(d)}_{m,n} = \int_0^{a^2} dt t^m e^{-\sigma t^{d/2}} \int_{\sqrt{t}}^a dv v^{n-m-1}. \tag{D21}$$

For $m \neq n$,

$$\int_{\sqrt{t}}^a dv v^{n-m-1} = \frac{a^{n-m} - t^{(n-m)/2}}{n-m}, \tag{D22}$$

and therefore

$$\mathcal{I}^{(d)}_{m,n} = \frac{1}{n-m}\left[a^{n-m}\int_0^{a^2} dt t^m e^{-\sigma t^{d/2}} - \int_0^{a^2} dt t^{(m+n)/2} e^{-\sigma t^{d/2}}\right]. \tag{D23}$$

Define

$$K_p(a,\sigma) := \int_0^{a^2} dt t^p e^{-\sigma t^{d/2}}. \tag{D24}$$

Using

$$z = \sigma t^{d/2}, \qquad t = \left(\frac{z}{\sigma}\right)^{2/d}, \qquad dt = \frac{2}{d}\sigma^{-2/d} z^{2/d-1} dz, \tag{D25}$$

we obtain

$$\begin{aligned} K_p(a,\sigma) &= \frac{2}{d}\sigma^{-2(p+1)/d}\int_0^{\sigma a^d} dz z^{2(p+1)/d-1} e^{-z} \\ &= \frac{2}{d}\sigma^{-2(p+1)/d}\gamma\left(\frac{2(p+1)}{d}, \sigma a^d\right). \end{aligned} \tag{D26}$$

Consequently, for $m \neq n$,

$$\mathcal{I}^{(d)}_{m,n}(a,\sigma) = \frac{2}{d(n-m)}\left[a^{n-m}\sigma^{-2(m+1)/d}\gamma\left(\frac{2(m+1)}{d}, \sigma a^d\right) - \sigma^{-(m+n+2)/d}\gamma\left(\frac{m+n+2}{d}, \sigma a^d\right)\right]. \tag{D27}$$

For the diagonal case $m = n$, the inner integral in (D21) becomes

$$\int_{\sqrt{t}}^{a} \frac{dv}{v} = \ln a - \frac{1}{2}\ln t. \tag{D28}$$

Thus,

$$\mathcal{I}^{(d)}_{m,m} = \int_0^{a^2} dt t^m e^{-\sigma t^{d/2}}\left(\ln a - \frac{1}{2}\ln t\right). \tag{D29}$$

Using

$$K_m(a,\sigma) = \frac{2}{d}\sigma^{-s_m}\gamma(s_m, \sigma a^d), \qquad s_m := \frac{2(m+1)}{d}, \tag{D30}$$

and

$$\int_0^{a^2} dt t^m \ln t e^{-\sigma t^{d/2}} = \frac{\partial K_m}{\partial m}, \tag{D31}$$

we find

$$\mathcal{I}^{(d)}_{m,m}(a,\sigma) = \frac{2}{d^2}\sigma^{-2(m+1)/d}\left[\gamma\left(\frac{2(m+1)}{d}, \sigma a^d\right)\ln(\sigma a^d) - \frac{\partial}{\partial s}\gamma\left(s, \sigma a^d\right)\Big|_{s=2(m+1)/d}\right]. \tag{D32}$$

For the large-σa^d limit, writing

$$\gamma(s, \sigma a^d) = \Gamma(s) - \Gamma(s, \sigma a^d), \tag{D33}$$

give us

$$\mathcal{I}^{(d)}_{m,n}(a,\sigma) = \frac{2}{d(n-m)}\left[a^{n-m}\Gamma\left(\frac{2(m+1)}{d}\right)\sigma^{-2(m+1)/d} - \Gamma\left(\frac{m+n+2}{d}\right)\sigma^{-(m+n+2)/d}\right] + \mathcal{O}\left(e^{-\sigma a^d}\right), \qquad m \neq n. \tag{D34}$$

and

$$\mathcal{I}^{(d)}_{m,m}(a,\sigma) = \frac{2}{d^2}\sigma^{-2(m+1)/d}\Gamma\left(\frac{2(m+1)}{d}\right)\left[\ln(\sigma a^d) - \psi\left(\frac{2(m+1)}{d}\right)\right] + \mathcal{O}\left(e^{-\sigma a^d}\right), \qquad m = n \tag{D35}$$

where

$$\psi(s) := \frac{d}{ds}\ln\Gamma(s) = \frac{\Gamma'(s)}{\Gamma(s)} \tag{D36}$$

is the digamma function.

For later use, it is convenient to identify explicitly the two asymptotic sectors in Eq. (D34):

$$\mathcal{I}^{(1)}_{m,n} = \frac{2a^{n-m}}{d(n-m)}\Gamma\left(\frac{2(m+1)}{d}\right)\sigma^{-2(m+1)/d}, \tag{D37}$$

$$\mathcal{I}^{(2)}_{m,n} = -\frac{2}{d(n-m)}\Gamma\left(\frac{m+n+2}{d}\right)\sigma^{-(m+n+2)/d}. \tag{D38}$$

The second sector is antisymmetric under interchange of the two indices:

$$\mathcal{I}^{(2)}_{m,n} = -\mathcal{I}^{(2)}_{n,m}, \qquad m \neq n. \tag{D39}$$

Consequently, a symmetric coefficient multiplying the pair (m,n) and (n,m) gives an exact cancellation of the second asymptotic sector. For the diagonal case,

$$\mathcal{I}^{(d)}_{m,m} \sim \sigma^{-2(m+1)/d}\left[A_m \ln\sigma + B_m\right], \tag{D40}$$

which is the origin of the logarithmic sectors discussed in Appendix A.

3. Euler-Operator Identity

Let

$$H := x\frac{d}{dx} \tag{D41}$$

denote the Euler operator, and let

$$P(t) = \sum_{k=0}^{m} c_k t^k \tag{D42}$$

be a polynomial. For $x > 0$ and $a \in \mathbb{C}$,

$$H(x^a) = a x^a, \tag{D43}$$

while

$$H\left(x^a \ln x\right) = a x^a \ln x + x^a. \tag{D44}$$

It follows by induction that

$$H^k\left(x^a \ln x\right) = a^k x^a \ln x + k a^{k-1} x^a, \tag{D45}$$

with the second term understood to vanish for $k = 0$. Hence, by linearity,

$$P(H)\left(x^a \ln x\right) = P(a) x^a \ln x + P'(a) x^a, \tag{D46}$$

where $P'(a)$ denotes the ordinary derivative of $P(t)$ with respect to its argument. In particular, if λ is a double root of the Euler polynomial, then

$$(D_\rho - \lambda)^2\left[\rho^\lambda (A\ln\rho + B)\right] = 0. \tag{D47}$$

A simple root removes the logarithm but leaves a term proportional to ρ^λ; a double root removes both.

4. Endpoint Values of the Euler Polynomial

For later use, we evaluate explicitly the values of the Euler polynomial that appear in Eqs. (A130) and (A135). From Eqs. (A101)-(A102),

$$Q_d(\lambda) = N_d\left(\lambda + \frac{d+2}{d}\right)\prod_{j=1}^{q_d}\left(\lambda + \frac{2j}{d}\right), \qquad q_d = \left\lfloor\frac{d+2}{2}\right\rfloor + 1 \tag{D48}$$

with

$$N_d = \frac{d^{q_d+1}}{2^{q_d} q_d!(d+2)} \tag{D49}$$

For odd $d = 2s+1$, one has $q_d = s+2$, and therefore

$$\begin{aligned} Q_d(-1) &= N_d \frac{2}{d}\prod_{j=1}^{s+2}\frac{2j-(2s+1)}{d} \\ &= \frac{3(-1)^s\Gamma\left(\frac{d}{2}\right)}{2\sqrt{\pi}(d+2)\Gamma\left(\frac{d+5}{2}\right)} \end{aligned} \tag{D50}$$

and

$$\begin{aligned} Q_d\left(-\frac{d+4}{d}\right) &= N_d\left(-\frac{2}{d}\right)\prod_{j=1}^{s+2}\frac{2j-(2s+5)}{d} \\ &= (-1)^{(d+5)/2}\frac{\Gamma\left(\frac{d+2}{2}\right)}{\sqrt{\pi}\Gamma\left(\frac{d+5}{2}\right)} \end{aligned} \tag{D51}$$

For even $d = 2s$, one has $q_d = s+2$, and both relevant arguments are roots of Q_d,

$$Q_d(-1) = 0, \qquad Q_d\left(-\frac{d+4}{d}\right) = 0 \tag{D52}$$

Their derivatives are obtained by removing the corresponding vanishing factor:

$$\begin{aligned} Q'_d(-1) &= N_d\frac{2}{d}\prod_{\substack{j=1\\ j\neq s}}^{s+2}\frac{j-s}{s} \\ &= \frac{16(-1)^{d/2-1}}{(d+2)^2(d+4)} \end{aligned} \tag{D53}$$

and

$$\begin{aligned} Q'_d\left(-\frac{d+4}{d}\right) &= N_d\left(-\frac{2}{d}\right)\prod_{j=1}^{s+1}\frac{j-s-2}{s} \\ &= (-1)^{d/2}\frac{2d}{(d+2)(d+4)} \end{aligned} \tag{D54}$$

These are the only values of Q_d and Q'_d required in the continuum-limit calculation.

5. Finite-Difference Inversion Formula

Let $F(x)$ be a polynomial of degree at most $n-1$. Introduce the discrete-binomial basis

$$e_i(x) := (-1)^{i-1}\binom{x}{i-1}, \qquad i = 1, \dots, n. \tag{D55}$$

Since $\{e_i(x)\}_{i=1}^n$ forms a basis for polynomials of degree at most $n-1$, we may write

$$F(x) = \sum_{i=1}^{n} C_i e_i(x). \tag{D56}$$

Define the forward-difference operator

$$\Delta f(x) := f(x+1) - f(x). \tag{D57}$$

Using

$$\Delta\binom{x}{k} = \binom{x}{k-1}, \tag{D58}$$

one obtains

$$\Delta^m e_i(0) = \begin{cases} (-1)^{i-1}, & m = i-1, \\ 0, & m \neq i-1. \end{cases} \tag{D59}$$

Applying Δ^{i-1} to Eq. (D56) and evaluating at $x = 0$ therefore isolates C_i:

$$C_i = (-1)^{i-1}\Delta^{i-1}F(0). \tag{D60}$$

The $(i-1)$ -th forward difference has the standard expansion

$$\Delta^{i-1}F(0) = \sum_{k=0}^{i-1}\binom{i-1}{k}(-1)^{i-1-k}F(k), \tag{D61}$$

and hence

$$C_i = \sum_{k=0}^{i-1}\binom{i-1}{k}(-1)^{k}F(k), \qquad i = 1, \dots, n. \tag{D62}$$

This inversion formula determines the coefficients of the discrete-binomial basis directly from the values of the polynomial at the integers $0, 1, \dots, n-1$.

6. Finite Binomial-Sum Identity

For $a \notin \{0, 1, \dots, n\}$, define

$$S_n(a) := \sum_{j=0}^{n} \frac{(-1)^j\binom{n}{j}}{a-j}. \tag{D63}$$

For $\mathrm{Re}(a) > n$, using

$$\frac{1}{a-j} = \int_0^1 t^{a-j-1}dt, \tag{D64}$$

and the binomial theorem,

$$S_n(a) = \int_0^1 dt t^{a-1}\sum_{j=0}^{n}(-1)^j\binom{n}{j}t^{-j} \tag{D65}$$

$$= (-1)^n\int_0^1 dt t^{a-n-1}(1-t)^n \tag{D66}$$

$$= (-1)^n B(a-n, n+1) \tag{D67}$$

$$= (-1)^n\frac{\Gamma(n+1)\Gamma(a-n)}{\Gamma(a+1)}. \tag{D68}$$

Both sides are meromorphic functions of a, and the result therefore extends by analytic continuation to

$$\sum_{j=0}^{n}\frac{(-1)^j\binom{n}{j}}{a-j} = (-1)^n\frac{\Gamma(n+1)\Gamma(a-n)}{\Gamma(a+1)} \tag{D69}$$

for $a \notin \{0, 1, \dots, n\}$. In particular, no restriction on the relative magnitudes of a and n is required.

7. Closed-Form Layer Coefficients

We now derive the layer coefficients appearing in Eq. (A3) directly from the Euler operator (A101). Let

$$q := q_d. \tag{D70}$$

The normalized density operator is

$$\widehat{\mathcal{E}}_d = \frac{d^{q+1}}{2^q q!(d+2)}\left(D_\rho + \frac{d+2}{d}\right)\prod_{j=1}^{q}\left(D_\rho + \frac{2j}{d}\right). \tag{D71}$$

To expand this operator in powers of D_ρ, introduce the unsigned Stirling numbers of the first kind, $\left[{n \atop k}\right]$, defined by

$$x^{\overline{n}} = x(x+1)\cdots(x+n-1) = \sum_{k=0}^{n}\left[{n \atop k}\right]x^k, \tag{D72}$$

where $x^{\overline{n}}$ denotes the rising factorial. They satisfy the recurrence relation

$$\left[{n+1 \atop k}\right] = \left[{n \atop k-1}\right] + n\left[{n \atop k}\right].$$

Since

$$\prod_{j=1}^{q}\left(D_\rho + \frac{2j}{d}\right) = \left(\frac{2}{d}\right)^q\prod_{j=1}^{q}\left(\frac{d}{2}D_\rho + j\right), \tag{D73}$$

we may write

$$\prod_{j=1}^{q}\left(D_\rho + \frac{2j}{d}\right) = \left(\frac{2}{d}\right)^q\left(\frac{d}{2}D_\rho + 1\right)^{\overline{q}}$$

$$= \left(\frac{2}{d}\right)^q\frac{\left(\frac{d}{2}D_\rho\right)^{\overline{q+1}}}{\frac{d}{2}D_\rho}$$

$$= \left(\frac{2}{d}\right)^q\sum_{r=0}^{q}\left[{q+1 \atop r+1}\right]\left(\frac{d}{2}\right)^r D_\rho^r. \tag{D74}$$

Therefore, defining

$$A_r^{(d)} := \frac{1}{q!}\begin{bmatrix} q+1 \\ r+1 \end{bmatrix}\left(\frac{d}{2}\right)^r, \qquad A_{-1}^{(d)} = A_{q+1}^{(d)} = 0, \tag{D75}$$

the normalized Euler operator becomes

$$\widehat{\mathcal{E}}_d = \sum_{r=0}^{q+1}\left[A_r^{(d)} + \frac{d}{d+2}A_{r-1}^{(d)}\right]D_\rho^r. \tag{D76}$$

We next determine the polynomial generated by this operator. Writing

$$z = \rho V, \tag{D77}$$

one has

$$D_\rho = z\frac{d}{dz}. \tag{D78}$$

The standard relation between powers of the Euler derivative and ordinary derivatives is

$$\left(z\frac{d}{dz}\right)^r = \sum_{m=0}^{r}\left\{ r \atop m \right\} z^m \frac{d^m}{dz^m}, \tag{D79}$$

where $\left\{ n \atop k \right\}$ are the Stirling numbers of the second kind.

The Stirling numbers of the second kind are defined by the expansion of the ordinary power in falling factorials,

$$x^n = \sum_{k=0}^{n}\left\{ n \atop k \right\} x^{\underline{k}},$$

where

$$x^{\underline{k}} := x(x-1)\cdots(x-k+1).$$

They obey the recurrence

$$\left\{ n+1 \atop k \right\} = \left\{ n \atop k-1 \right\} + k\left\{ n \atop k \right\}.$$

Since

$$\frac{d^m}{dz^m}e^{-z} = (-1)^m e^{-z}, \tag{D80}$$

it follows that

$$D_\rho^r e^{-z} = e^{-z}\sum_{m=0}^{r}(-1)^m\left\{ r \atop m \right\} z^m. \tag{D81}$$

Applying Eq. (D76) therefore yields

$$\widehat{\mathcal{E}}_d e^{-z} = e^{-z}\sum_{m=0}^{q+1}(-1)^m\left[\sum_{r=m}^{q+1}\left(A_r^{(d)} + \frac{d}{d+2}A_{r-1}^{(d)}\right)\left\{ r \atop m \right\}\right] z^m. \tag{D82}$$

Comparing this with

$$\mathcal{P}_d(z)e^{-z} = \sum_{i=1}^{q+2} C_i^{(d)}\frac{z^{i-1}}{(i-1)!}e^{-z}, \tag{D83}$$

and setting $m = i-1$, gives the closed-form expression

$$C_i^{(d)} = (-1)^{i-1}(i-1)!\sum_{r=i-1}^{q+1}\left[A_r^{(d)} + \frac{d}{d+2}A_{r-1}^{(d)}\right]\left\{ r \atop i-1 \right\}, \tag{D84}$$

where $i = 1, \ldots, q+2$. Equivalently,

$$C_i^{(d)} = \frac{(-1)^{i-1}(i-1)!}{q!}\sum_{r=i-1}^{q+1}\left[\begin{bmatrix} q+1 \\ r+1 \end{bmatrix}\left(\frac{d}{2}\right)^r + \frac{d}{d+2}\begin{bmatrix} q+1 \\ r \end{bmatrix}\left(\frac{d}{2}\right)^{r-1}\right]\left\{ r \atop i-1 \right\}. \tag{D85}$$